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Qualification Pack

Junior Engineer - Rolling Stock (Metro Rail)

QP Code: DMR/LSC/Q0301

Version: 1.0

NSQF Level: 4.5

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DMR/LSC/Q0301: Junior Engineer - Rolling Stock (Metro Rail)

Brief Job Description

A Junior Engineer - Rolling Stock (Metro Rail) is responsible for overseeing the maintenance and safety compliance of various rolling stock systems. It involves conducting preventive maintenance checks, implementing essential safety and maintenance procedures, and ensuring that rolling stock operates optimally. The individual will also participate in GMSM and MSM simulator training for enhanced diagnostic skills and engage in specialized training across traction, electrical & mechanical (E&M), signaling & telecommunications (S&T), P-way, and related departments to maintain proficiency. An understanding of VCC and pneumatic circuit operations is critical to support safe, efficient rolling stock performance.

Personal Attributes

The individual should be detail-oriented, safety-conscious, and adept at troubleshooting complex systems. Strong organizational skills, proactive problem-solving, and effective communication are essential for ensuring smooth operations and team coordination.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. DMR/LSC/N9901: Implement metro operations training and safety procedures
2. DMR/LSC/N0301: Understand and Compare Rolling Stock and Other Rolling Stock Systems,Safety, and M &P Fundamentals
3. DMR/LSC/N0302: Develop Proficiency in Other Rolling Stock Systems
4. DMR/LSC/N0303: Implement Key Safety and Maintenance Procedures for Rolling Stock
5. DMR/LSC/N0304: Conduct Preventive Maintenance Checks for Rolling Stock
6. DMR/LSC/N0305: Undergo GMSM & MSM Maintenance Simulator Training
7. DMR/LSC/N0306: Undergo Training on Important VCC & Pneumatic Circuits of Rolling Stock and Other Rolling Stocks
8. DGT/VSQ/N0101: Employability Skills (30 Hours)

Qualification Pack (QP) Parameters

Sector	Logistics

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GOVERNMENT OF INDIA MINISTRY OF SKILL DEVELOPMENT & ENTREPRENEURSHIP

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Qualification Pack

Sub-Sector	Transport (Metro Rail)
Occupation	Rolling Stock
Country	India
NSQF Level	4.5
Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2153.9900
Minimum Educational Qualification & Experience	Diploma (3 years Engineering Diploma / equivalent in Electrical / Mechanical / Electronics or in relevant field)
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	18/02/2028
NSQC Approval Date	18/02/2025
Version	1.0
Reference code on NQR	QG-4.5-TW-03513-2025-V
NQR Version	1.0

DMR/LSC/N9901: Implement metro operations training and safety procedures

Description

This unit focuses on implementing structured training and safety measures for metro operations, ensuring technical knowledge, system readiness, and emergency preparedness.

Scope

The scope covers the following :

- Training Module Overview
- Safety Overview
- Store Management and Procurement
- Overview of Telecom, Signaling, and Track Systems
- Overview of Traction, SCADA, and E&M Systems
- OCC and Rolling Stock Overview
- Operations and Customer Care
- Practical Visits
- First Aid Training
- Fire Fighting Systems
- Security in Metro Systems
- Metro Rail General Rules 2020
- Safety Circulars and Contingency Plans

Elements and Performance Criteria

Training Module Overview

To be competent, the user/individual on the job must be able to:

PC1. follow schedules for weekly activities and guidelines on punctuality, discipline, and conduct.

PC2. conduct identity verification and maintain attendance records

PC3. ensure adherence to training timelines and program deliverables

Safety Overview

To be competent, the user/individual on the job must be able to:

PC4. identify the role of the safety department and follow measures for personal, passenger, and system safety

PC5. use appropriate Personal Protective Equipment (PPE) in all designated areas

PC6. follow the hazard identification and risk assessment processes

Store Management and Procurement

To be competent, the user/individual on the job must be able to:

PC7. perform store management activities, including stock handling, documentation, and interdepartmental coordination for procurement.

PC8. monitor inventory levels and ensure timely requisitions for required materials

PC9. maintain records of procurement processes and vendor communication

Overview of Telecom, Signaling, and Track Systems

To be competent, the user/individual on the job must be able to:

PC10. identify telecom components (AFC, CCTV, PIDS, radio systems, clocks) and perform basic functional checks

PC11. identify signaling systems (CATC, CBTC) and observe/report signal functioning during visits.

PC12. identify P-Way components (rails, sleepers, crossings) and report track conditions through basic patrolling tasks

PC13. report malfunctions in telecom, signaling, or track systems using the correct reporting protocols

Overview of Traction, SCADA, and E&M Systems

To be competent, the user/individual on the job must be able to:

PC14. identify traction power and SCADA components and interpret sectioning diagrams

PC15. identify E&M systems (RSS, ASS, DG sets, UPS) and perform basic checks to understand their functionality

PC16. report unusual system performance and assist in routine inspections

OCC and Rolling Stock Overview

To be competent, the user/individual on the job must be able to:

PC17. familiarize with OCC systems, tools, and the role of OCC controllers

PC18. identify rolling stock components and their maintenance procedures

PC19. report rolling stock irregularities observed during operations

Operations and Customer Care

To be competent, the user/individual on the job must be able to:

PC20. familiarize with station operations, revenue management, and crew control functions

PC21. identify public complaint channels, possible causes, and resolve complaints using proper processes

PC22. assist in managing passenger inquiries and ensuring customer satisfaction

Practical Visits

To be competent, the user/individual on the job must be able to:

PC23. conduct visits to OCC, stations, depots, and tunnels to observe layouts, facilities, and operations

PC24. document key observations during practical visits for knowledge reinforcement

PC25. identify operational challenges during visits and suggest improvement areas

First Aid Training

To be competent, the user/individual on the job must be able to:

PC26. identify the role and scope of first aid, including key human body functions (e.g., circulatory, respiratory systems)

PC27. follow appropriate techniques for treating bleeding, fractures, burns, shock, and unconsciousness

PC28. perform artificial respiration, CPR, and safe transportation of injured persons using stretchers

PC29. participate in simulated first aid scenarios to enhance preparedness

Fire Fighting Systems

To be competent, the user/individual on the job must be able to:

PC30. identify elements of combustion, classes of fire, and appropriate extinguishing methods

PC31. operate active/passive fire protection systems, including alarms, fire extinguishers, and fire hydrants

PC32. conduct practical drills and identify fire suppression equipment at metro stations

PC33. respond to fire incidents promptly, ensuring passenger and system safety

Security in Metro Systems

To be competent, the user/individual on the job must be able to:

PC34. identify security threats, red alert situations, and follow protocols for disaster management

PC35. handle theft cases, manage abandoned persons, and lodge FIRs as per protocol

PC36. coordinate with security personnel to mitigate security breaches effectively

Metro Rail General Rules 2020

To be competent, the user/individual on the job must be able to:

PC37. interpret key provisions under Metro Rail General Rules 2020, including signals, train speed limits, and safety protocols

PC38. follow operational procedures for maintenance, station control duties, and driverless train systems

PC39. ensure compliance with rules for safe and efficient train operations

Safety Circulars and Contingency Plans

To be competent, the user/individual on the job must be able to:

PC40. implement procedures for the imposition of TSR, axle counter reset, Train ID management, and Computer Aided Train Supervision (CATS) operation

PC41. follow guidelines for engineering possession and maintenance activities during non-revenue hours, including hot work, solar system testing/repair, and role-play exercises

PC42. ensure compliance with procedures for stopping trains at platforms and during operational disruptions, such as earthquakes, high winds, and floods

PC43. manage degraded mode operations, including train movement during interlocking failures

PC44. oversee the issue, revision, updating, and deletion of SOPs, SWO/DWO, and timetable trials for new sections

PC45. facilitate the initiation of revenue train services, including pilot train movement, ATO/UTO/ATB mode operations, and AFTC functionality checks

PC46. execute procedures for negotiating neutral sections, managing pantograph failures, and monitoring Panto-OHE interface cameras

PC47. oversee train stabling, depot entry/exit, train movement, and fitness checks

PC48. conduct emergency evacuation using UG facilities, mid-tunnel shafts, and walkways for detrainment of passengers due to technical failure, explosion, or fire

PC49. operate and manage non-ATP vehicles such as RGM and CMV/CTMC and address incidents like rail fractures, weld failures, rollback scenarios, radio communication issues and resetting of passenger alarms in train

PC50. coordinate vehicle movement on link lines and rescue immobilized trains

PC51. respond to fire/smoke incidents, including UG train fires, ensuring passenger and staff safety

PC52. operate ETS/BLS plungers, remove foreign objects like kites from OHE, and follow OHE safety guidelines

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the importance of punctuality, discipline, and structured training
KU2. the procedures for trainee verification and attendance record maintenance
KU3. the role of time management in achieving training milestones
KU4. the role of the safety department and station/depot safety protocols
KU5. the use and importance of PPE
KU6. the basics of hazard identification, risk management, and accident reporting
KU7. the basics of stores management (inventory control, documentation)
KU8. the procurement processes and coordination across departments
KU9. the procedures for vendor management and maintaining procurement transparency
KU10. the telecom components, including AFC, CCTV, clocks, and communication systems
KU11. the basics of signaling systems (CATC, CBTC) and reporting unusual signal observations
KU12. the key P-Way components, track patrolling, and reporting track conditions
KU13. the common malfunctions in telecom, signaling, and track systems
KU14. the traction power systems and sectioning diagrams
KU15. the importance of SCADA and its functionalities
KU16. the role of E&M systems (DG sets, UPS) and their use during emergencies
KU17. the key preventive maintenance checks for E&M systems
KU18. the OCC systems, tools, and controller responsibilities
KU19. the rolling stock components, features, and basic maintenance
KU20. the reporting and addressing of common rolling stock issues
KU21. the station operations, revenue management, and crew coordination
KU22. the causes of public complaints and grievance resolution processes
KU23. the techniques for effective customer service and passenger engagement
KU24. the observations and learning outcomes from OCC, station, depot, and tunnel visits
KU25. the identifying real-world challenges during visits and linking theory to practice
KU26. the basic principles of first aid and key human body functions
KU27. the management of bleeding, fractures, burns, and unconsciousness
KU28. the artificial respiration, CPR, and safe patient transportation
KU29. the importance of first aid preparedness during emergencies
KU30. the elements of combustion, fire classifications, and extinguishing methods
KU31. the principles of passive/active fire protection systems
KU32. the practical usage of fire extinguishers, hydrants, and drills
KU33. the emergency response procedures during fire incidents
KU34. the security protocols, red alert situations, and theft management processes
KU35. the protocols for managing abandoned persons and disaster situations
KU36. the role of security personnel in ensuring passenger and asset safety
KU37. the key terminologies, signal usage, and operational protocols

KU38. the rules for maintenance works, train operations, and driverless train systems
KU39. the role of regulatory compliance in metro rail safety
KU40. the safety circulars and contingency plans for fire/security incidents
KU41. the importance of proactive contingency planning for emergency scenarios.
KU42. the importance role-playing and drills to improve safety and response performance
KU43. the procedures for TSR imposition, axle counter reset, Train ID management, and CATS operations
KU44. the engineering possession protocols for maintenance tasks during non-revenue hours, including solar system testing and hot work
KU45. the guidelines for safely halting train operations during emergencies, natural disasters, or technical failures
KU46. the steps for managing degraded mode operations and interlocking failures
KU47. the SOP and SWO/DWO lifecycle management, including revisions and timetable trials for new sections
KU48. the ATO, UTO, and ATB mode operations, emphasizing operational readiness and AFTC checks
KU49. the procedures for neutral section negotiation, pantograph damage management, and monitoring Panto-OHE interfaces using cameras
KU50. the best practices for train stabling, depot operations, and train fitness verification
KU51. the emergency evacuation protocols, focusing on passenger safety in UG and mid-tunnel situations
KU52. the guidelines for operating non-ATP vehicles, handling rail fractures, weld failures, and resetting alarms during rollbacks
KU53. the vehicle operations on link lines and immobilized train rescue procedures
KU54. the protocols for fire/smoke incidents, particularly UG fires, and ensuring safety during such incidents
KU55. the operation of ETS/BLS plungers and removal of foreign objects from OHE systems, along with OHE safety practices

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. convey clear and concise information to teams, junior engineers, and customers
GS2. effectively prioritize tasks and adhere to training schedules and operational timelines
GS3. identify operational challenges and propose practical solutions during training or real-time operations
GS4. assess system performance, identify malfunctions, and report anomalies accurately
GS5. work cohesively with peers, instructors, and junior engineers to meet training objectives
GS6. address passenger queries and manage grievances professionally
GS7. proactively identify risks, follow safety protocols, and contribute to hazard-free operations
GS8. adapt to diverse training environments, system layouts, and evolving operational scenarios
GS9. accurately maintain logs, records, and reports for training and practical activities

GS10. monitor systems, equipment, and processes, identifying areas requiring intervention or improvement

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Training Module Overview	**2**	**2**	**-**	**1**
PC1. follow schedules for weekly activities and guidelines on punctuality, discipline, and conduct.	-	-	-	-
PC2. conduct identity verification and maintain attendance records	-	-	-	-
PC3. ensure adherence to training timelines and program deliverables	-	-	-	-
Safety Overview	**2**	**2**	**-**	**1**
PC4. identify the role of the safety department and follow measures for personal, passenger, and system safety	-	-	-	-
PC5. use appropriate Personal Protective Equipment (PPE) in all designated areas	-	-	-	-
PC6. follow the hazard identification and risk assessment processes	-	-	-	-
Store Management and Procurement	**2**	**2**	**-**	**1**
PC7. perform store management activities, including stock handling, documentation, and interdepartmental coordination for procurement.	-	-	-	-
PC8. monitor inventory levels and ensure timely requisitions for required materials	-	-	-	-
PC9. maintain records of procurement processes and vendor communication	-	-	-	-
Overview of Telecom, Signaling, and Track Systems	**2**	**2**	**-**	**1**
PC10. identify telecom components (AFC, CCTV, PIDS, radio systems, clocks) and perform basic functional checks	-	-	-	-
PC11. identify signaling systems (CATC, CBTC) and observe/report signal functioning during visits.	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. identify P-Way components (rails, sleepers, crossings) and report track conditions through basic patrolling tasks	-	-	-	-
PC13. report malfunctions in telecom, signaling, or track systems using the correct reporting protocols	-	-	-	-
Overview of Traction, SCADA, and E&M Systems	**2**	**2**	**-**	**1**
PC14. identify traction power and SCADA components and interpret sectioning diagrams	-	-	-	-
PC15. identify E&M systems (RSS, ASS, DG sets, UPS) and perform basic checks to understand their functionality	-	-	-	-
PC16. report unusual system performance and assist in routine inspections	-	-	-	-
OCC and Rolling Stock Overview	**2**	**2**	**-**	**1**
PC17. familiarize with OCC systems, tools, and the role of OCC controllers	-	-	-	-
PC18. identify rolling stock components and their maintenance procedures	-	-	-	-
PC19. report rolling stock irregularities observed during operations	-	-	-	-
Operations and Customer Care	**2**	**2**	**-**	**1**
PC20. familiarize with station operations, revenue management, and crew control functions	-	-	-	-
PC21. identify public complaint channels, possible causes, and resolve complaints using proper processes	-	-	-	-
PC22. assist in managing passenger inquiries and ensuring customer satisfaction	-	-	-	-
Practical Visits	**1**	**1**	**-**	**0.5**
PC23. conduct visits to OCC, stations, depots, and tunnels to observe layouts, facilities, and operations	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. document key observations during practical visits for knowledge reinforcement	-	-	-	-
PC25. identify operational challenges during visits and suggest improvement areas	-	-	-	-
First Aid Training	**1**	**1**	**-**	**0.5**
PC26. identify the role and scope of first aid, including key human body functions (e.g., circulatory, respiratory systems)	-	-	-	-
PC27. follow appropriate techniques for treating bleeding, fractures, burns, shock, and unconsciousness	-	-	-	-
PC28. perform artificial respiration, CPR, and safe transportation of injured persons using stretchers	-	-	-	-
PC29. participate in simulated first aid scenarios to enhance preparedness	-	-	-	-
Fire Fighting Systems	**1**	**1**	**-**	**0.5**
PC30. identify elements of combustion, classes of fire, and appropriate extinguishing methods	-	-	-	-
PC31. operate active/passive fire protection systems, including alarms, fire extinguishers, and fire hydrants	-	-	-	-
PC32. conduct practical drills and identify fire suppression equipment at metro stations	-	-	-	-
PC33. respond to fire incidents promptly, ensuring passenger and system safety	-	-	-	-
Security in Metro Systems	**1**	**1**	**-**	**0.5**
PC34. identify security threats, red alert situations, and follow protocols for disaster management	-	-	-	-
PC35. handle theft cases, manage abandoned persons, and lodge FIRs as per protocol	-	-	-	-
PC36. coordinate with security personnel to mitigate security breaches effectively	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Metro Rail General Rules 2020	**1**	**1**	**-**	**0.5**
PC37. interpret key provisions under Metro Rail General Rules 2020, including signals, train speed limits, and safety protocols	-	-	-	-
PC38. follow operational procedures for maintenance, station control duties, and driverless train systems	-	-	-	-
PC39. ensure compliance with rules for safe and efficient train operations	-	-	-	-
Safety Circulars and Contingency Plans	**1**	**1**	**-**	**0.5**
PC40. implement procedures for the imposition of TSR, axle counter reset, Train ID management, and Computer Aided Train Supervision (CATS) operation	-	-	-	-
PC41. follow guidelines for engineering possession and maintenance activities during non-revenue hours, including hot work, solar system testing/repair, and role-play exercises	-	-	-	-
PC42. ensure compliance with procedures for stopping trains at platforms and during operational disruptions, such as earthquakes, high winds, and floods	-	-	-	-
PC43. manage degraded mode operations, including train movement during interlocking failures	-	-	-	-
PC44. oversee the issue, revision, updating, and deletion of SOPs, SWO/DWO, and timetable trials for new sections	-	-	-	-
PC45. facilitate the initiation of revenue train services, including pilot train movement, ATO/UTO/ATB mode operations, and AFTC functionality checks	-	-	-	-
PC46. execute procedures for negotiating neutral sections, managing pantograph failures, and monitoring Panto-OHE interface cameras	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC47. oversee train stabling, depot entry/exit, train movement, and fitness checks	-	-	-	-
PC48. conduct emergency evacuation using UG facilities, mid-tunnel shafts, and walkways for detrainment of passengers due to technical failure, explosion, or fire	-	-	-	-
PC49. operate and manage non-ATP vehicles such as RGM and CMV/CTMC and address incidents like rail fractures, weld failures, rollback scenarios, radio communication issues and resetting of passenger alarms in train	-	-	-	-
PC50. coordinate vehicle movement on link lines and rescue immobilized trains	-	-	-	-
PC51. respond to fire/smoke incidents, including UG train fires, ensuring passenger and staff safety	-	-	-	-
PC52. operate ETS/BLS plungers, remove foreign objects like kites from OHE, and follow OHE safety guidelines	-	-	-	-
NOS Total	**20**	**20**	**-**	**10**

National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N9901
NOS Name	Implement metro operations training and safety procedures
Sector	Logistics
Sub-Sector	
Occupation	Foundation – Metro (Rail)
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

DMR/LSC/N0301: Understand and Compare Rolling Stock and Other Rolling Stock Systems,Safety, and M &P Fundamentals

Description

This unit covers the foundational knowledge of Rolling Stock vehicle systems with a comparative overview of other rolling stock, covering key machinery and plant (M&P) operations along with essential safety protocols for railway systems.

Scope

The scope covers the following :

- Rolling Stock System Training and Comparison with Other Rolling Stock
- Practical Training on Rolling Stock System
- Practical Training on other Rolling Stock System & Depot
- Tools and Test Equipment
- Machinery and Plant (M&P)
- Stores
- Safety Precautions
- Maintenance Philosophy
- Budget Management
- Organizational Chart
- 21-Hour Reports and T&Q Cell Reporting
- Section Responsibilities
- Rolling Stock Comparison
- Depot & M&Ps
- Depot Control Center (DCC)
- Rolling Stock Care (RSC)

Elements and Performance Criteria

Rolling Stock System Training and Comparison with other rolling stock

To be competent, the user/individual on the job must be able to:

PC1. perform a detailed overview of Rolling Stock and Other Rolling Stocks types of trains, configurations, safety features, salient features, and various systems

PC2. carry out equipment layout checks, including saloon interior and exterior configurations, cab equipment, different panels, switches, push buttons, CBs, locations, and functions

PC3. inspect bogie and wheels, gangway, and coupler setups by identifying and comparing different parts and equipment

PC4. assess brakes and pneumatic systems, identifying brake types, their purpose and functions, various pneumatic valves, compressors, reservoirs, and piping colour codes for both stocks

PC5. inspect door system in Rolling Stock and Other Rolling Stocks, covering different door types, their purposes and functions, and the location of various equipment associated with the doors

PC6. check the saloon and cab air conditioning systems in Rolling Stock and Other Rolling Stocks, including major components, refrigeration cycle, different modes of operation, and temperature settings

PC7. assess propulsion system, covering high-tension and propulsion systems like pantograph, air feed insulators, VCB, PT, CT, EGS, main transformer, CI, and traction motors with their locations and functions

PC8. evaluate the auxiliary power supply system, examining SIV and battery modules, components, loads, and ratings, identifying key differences in location and function

PC9. examine the PA/PIS system's major systems, subsystems, modules like AOP, MOP, PAMP, displays, and push buttons, ensuring their proper functioning across Rolling Stock and Other Rolling Stocks

PC10. inspect TIMS in Rolling Stock and Other Rolling Stocks, verifying modules fitted in different cars with redundancy, and reviewing various modes/screens on Video Display Units

Practical Training on Rolling Stock System

To be competent, the user/individual on the job must be able to:

PC11. check the Rolling Stock equipment layout and functions, including saloon interior and exterior, under-frame, and roof equipment, ensuring all components are located and identified

PC12. identify cab equipment including different panels, switches, push buttons, CBs, their locations, and functions, for thorough understanding to perform tasks independently

PC13. inspect and assess bogie and wheels, gangway, and coupler systems by identifying different parts and equipment, along with their purpose and function

PC14. identify different types of brakes and their respective purposes and functions, including various pneumatic valves, compressors, reservoirs, and colour-coded piping

PC15. identify different doors types, their purposes and functions, and the location of equipment associated with doors

PC16. train on the saloon and cab AC, identifying major components, explaining the refrigeration cycle, and applying different modes of operation and temperature settings

PC17. operate the propulsion system, covering pantograph (Panto), vacuum circuit breaker (VCB), main transformer (MTR), control inverter (CI), and traction motors (TM)

PC18. carry out operational and functional checks on the auxiliary power supply system, covering major components within the Static Inverter (SIV) and battery systems

PC19. train on the TIMS and PA/PIS System, handling key operations, including modules like the AOP, MOP, PAMP, displays, and push buttons

Practical Training on other Rolling Stock System & Depot

To be competent, the user/individual on the job must be able to:

PC20. train on saloon and cab equipment in other rolling stock stocks, covering different panels, switches, push buttons, and CBS, identifying their locations, functions and differences compared to rolling stock

PC21. inspect other rolling stock bogie and wheels, gangway, coupler, brakes, and pneumatic systems, applying practical knowledge of different parts, equipment functions, and differences from rolling stock

PC22. inspect other rolling stock door systems, identifying various door types, their purposes, functions, and associated equipment locations, while understanding operational differences from rolling stock

PC23. train on other rolling stock saloon and cab ac units, applying knowledge of major components, the refrigeration cycle, and temperature setting modes with noted differences from rolling stock

PC24. train on TIMS and PA/PIS in other rolling stock, handling key modules and operations such as displays, push buttons, and announcements, focusing on the differences from rolling stock systems

PC25. operate the other rolling stock propulsion system, including Panto, VCB, MTR, CI, and traction motors, ensuring familiarity with each component's location, function, and any differences from rolling stock

PC26. carry out operational checks on other rolling stock auxiliary power supply systems, including SIV and battery systems, identifying different loads, modules, and components, noting differences from rolling stock

Tools and Test Equipment

To be competent, the user/individual on the job must be able to:

PC27. identify and select appropriate tools and test equipment based on their type, function, parameters, and purpose, ensuring their accurate application for maintenance and testing tasks in line with specific requirements

PC28. calibrate, and utilize tools and test equipment following manufacturer guidelines, technical standards, and established calibration procedures to ensure proper handling, reliability, and accuracy

Machinery and Plant (M&P)

To be competent, the user/individual on the job must be able to:

PC29. identify, inspect, and operate depot M&P equipment, ensuring readiness, efficiency, and safe usage for maintenance tasks

PC30. perform routine checks on depot M&P and report issues promptly to ensure smooth operations and minimize downtime

Stores

To be competent, the user/individual on the job must be able to:

PC31. oversee custody stores, tool rooms, and DCOS, managing stock and non-stock items, consumables, and tools by maintaining organized records, overseeing issuance and returns, and ensuring operational readiness

PC32. forecast and monitor inventory requirements, including EAC and PAC, by conducting stock verification, implementing rotation practices, auditing compliance, and tracking budgets to minimize wastage and ensure efficient resource utilization

Safety Precautions

To be competent, the user/individual on the job must be able to:

PC33. implement and enforce safety protocols in the Depot Yard, IBL, SBL, on roofs, near M&Ps, and OHE areas, including use of PPEs for various tasks

PC34. apply specific safety measures when working on trains, welding, and electrical maintenance, ensuring proper isolation, grounding, and protective gear

PC35. follow safe procedures for train coupling/uncoupling and shunting, adhering to alignment, signaling, and safe distance guidelines

PC36. execute safety checks for lifting operations and movement of heavy items, including use of overhead cranes, lifting gears, and proper techniques

PC37. ensure compliance with M&P safety protocols for equipment like Pit Jacks, Mobile Jacks, PWL, AWP, and RRV, covering secure setup and load limits

Maintenance Philosophy

To be competent, the user/individual on the job must be able to:

PC38. distinguish between preventive, corrective, and breakdown maintenance, applying each as appropriate

PC39. document and report on maintenance activities, ensuring efficient resource use

Budget Management

To be competent, the user/individual on the job must be able to:

PC40. prepare and manage budgets for rolling stock maintenance, including allocation for consumables and stock items

PC41. monitor budget utilization and optimize expenditures in rolling stock maintenance

Organizational Chart

To be competent, the user/individual on the job must be able to:

PC42. familiarize staff with the organization chart and functional roles in the RS depot and CPTD Cell

PC43. facilitate collaboration among sections such as PPIO, INSP, TECH CELL, and FIT

21-Hour Reports and T&Q Cell Reporting

To be competent, the user/individual on the job must be able to:

PC44. analyze 21-hour failure reports and other T&Q Cell documents to identify trends and recurring failures

PC45. analyze 21-hour failure reports and other T&Q Cell documents to identify trends and recurring failures

Section Responsibilities

To be competent, the user/individual on the job must be able to:

PC46. familiarize with roles and responsibilities of sections like PPIO, INSP, TECH CELL, FIT, MW, RSC, and DCC

PC47. coordinate efficient functioning within and between sections for seamless RS operations

Rolling Stock Comparison

To be competent, the user/individual on the job must be able to:

PC48. distinguish between all rolling stock types, identifying unique features and maintenance requirements

PC49. train the relevant personnel on stock-specific maintenance practices and equipment handling protocols

Depot & M&Ps

To be competent, the user/individual on the job must be able to:

PC50. train on rolling stock depot layout, covering the depot’s design, functional zones, and workflow to ensure effective navigation and operational efficiency within the depot environment

PC51. train on important M&Ps in rolling stock depots, including operation, maintenance procedures, and safety protocols specific to critical M&P equipment

Depot Control Center (DCC)

To be competent, the user/individual on the job must be able to:

PC52. analyze the DCC working to understand the full operational flow, including monitoring, coordination, and communication functions essential for depot operations

PC53. determine the role and tasks of the Points man within the DCC framework, e.g. managing track switches, ensuring safe movement of trains within the depot, etc

Rolling Stock Care (RSC)

To be competent, the user/individual on the job must be able to:

PC54. conduct RSC visits to inspect and ensure proper rolling stock care

PC55. report on RSC findings and coordinate with depot staff to address any identified issues

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. metro train types, configurations, safety features, key specifications, and systems

KU2. the rolling stock and other rolling stocks equipment layout and overview of saloon interiors, exteriors, cab equipment, panels, switches, push buttons, and CB locations with functions

KU3. the rolling stock and other rolling stocks bogies, wheels, gangways, and couplers, detailing parts, equipment purposes, and functions

KU4. the rolling stock and other rolling stocks brakes and pneumatic systems, brake types, pneumatic valves, compressors, reservoirs, and piping with colour codes

KU5. the rolling stock and other rolling stocks door systems, door types with purposes and functions, and equipment locations

KU6. the rolling stock and other rolling stocks air conditioning systems (saloon and cab), components, refrigeration cycle, operational modes, and temperature settings

KU7. the propulsion systems (Pantograph, VCB, MTR, CI, TM), high tension, and subsystems like pantographs, air feed insulators, VCB, PT, CT, EGS, main transformers, CI, and traction motors with locations and functions

KU8. the rolling stock and other rolling stocks auxiliary power supply systems (SIV & Battery), components, locations, functions, loads, and ratings

KU9. the rolling stock and other rolling stocks PA/PIS systems and subsystems, including AOP, MOP, PAMP, displays, and push buttons

KU10. the Train Information Management System (TIMS), modules by car type, redundancy, modes/screens on video display units

KU11. TIMS and PA/PIS in other rolling stock vs. rolling stock

KU12. the use of tools and test equipment, including parameters, purposes, uses, specifications, and calibration

KU13. the overview of Depot M&P

KU14. the overview of custody stores, tool room, DCOS, stock and non-stock items, consumables, EAC, PAC, and budget

KU15. the safety overview and do's & don'ts for depot yard, IBL, SBL, roofs, M&Ps, OHE, and PPE use

KU16. the safety protocols for train work, welding, electrical maintenance, coupling, uncoupling, and shunting

KU17. safety precautions for handling lifting gear, moving heavy items, using overhead cranes, and working on depot M&Ps (Pit Jacks, Mobile Jacks, PWL, AWP, RRV)
KU18. the maintenance philosophies, distinguishing preventive, corrective, and breakdown maintenance with examples
KU19. overview of budget planning
KU20. the organizational chart of RS and depot working, including CPTD Cell functions
KU21. Twenty one hour failure reports and other T&Q Cell reports
KU22. the roles and responsibilities of PPIO, INSP, TECH CELL, FIT, MW, RSC, and DCC sections
KU23. the differences among all train stocks
KU24. rolling stock depot layout
KU25. the key M&Ps in rolling stock depots
KU26. DCC operations, including pointsman duties
KU27. RSC visit overview

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. communicate effectively with team members, report findings, and explain technical aspects of rolling stock and other rolling stocks systems, ensuring clear and concise information exchange
GS2. maintain accurate records, such as equipment inspection reports, maintenance logs, and safety compliance documents, ensuring proper documentation of tasks and system performance
GS3. anticipate potential issues in the rolling stock and other rolling stocks systems, identify faults, and take preventive measures
GS4. resolve technical challenges through systematic troubleshooting techniques
GS5. make informed decisions based on system assessments, operational requirements, and safety protocols, ensuring efficient operations and adherence to safety standards
GS6. prioritize and manage time efficiently when inspecting systems, performing maintenance tasks, and conducting safety checks to meet deadlines and operational goals
GS7. collaborate effectively with colleagues and cross-functional teams (e.g., PPIO, TECH CELL) to ensure smooth operations and maintenance of Other Rolling Stock systems
GS8. focus attention on identifying system discrepancies, inspecting components, and ensuring all equipment is functioning as required
GS9. assess various components of rolling stock and other rolling stocks systems, analyze their functionality, and compare different configurations to optimize system performance and ensure safety
GS10. adjust to new tasks, systems, or technologies related to Other Rolling Stock systems, ensuring that any changes or updates to the system are efficiently handled

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Rolling Stock System Training and Comparison with other rolling stock	**2**	**2**	**-**	**0.5**
PC1. perform a detailed overview of Rolling Stock and Other Rolling Stocks types of trains, configurations, safety features, salient features, and various systems	-	-	-	-
PC2. carry out equipment layout checks, including saloon interior and exterior configurations, cab equipment, different panels, switches, push buttons, CBs, locations, and functions	-	-	-	-
PC3. inspect bogie and wheels, gangway, and coupler setups by identifying and comparing different parts and equipment	-	-	-	-
PC4. assess brakes and pneumatic systems, identifying brake types, their purpose and functions, various pneumatic valves, compressors, reservoirs, and piping colour codes for both stocks	-	-	-	-
PC5. inspect door system in Rolling Stock and Other Rolling Stocks, covering different door types, their purposes and functions, and the location of various equipment associated with the doors	-	-	-	-
PC6. check the saloon and cab air conditioning systems in Rolling Stock and Other Rolling Stocks, including major components, refrigeration cycle, different modes of operation, and temperature settings	-	-	-	-
PC7. assess propulsion system, covering high-tension and propulsion systems like pantograph, air feed insulators, VCB, PT, CT, EGS, main transformer, CI, and traction motors with their locations and functions	-	-	-	-
PC8. evaluate the auxiliary power supply system, examining SIV and battery modules, components, loads, and ratings, identifying key differences in location and function	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. examine the PA/PIS system's major systems, subsystems, modules like AOP, MOP, PAMP, displays, and push buttons, ensuring their proper functioning across Rolling Stock and Other Rolling Stocks	-	-	-	-
PC10. inspect TIMS in Rolling Stock and Other Rolling Stocks, verifying modules fitted in different cars with redundancy, and reviewing various modes/screens on Video Display Units	-	-	-	-
Practical Training on Rolling Stock System	**1**	**1**	**-**	**0.5**
PC11. check the Rolling Stock equipment layout and functions, including saloon interior and exterior, under-frame, and roof equipment, ensuring all components are located and identified	-	-	-	-
PC12. identify cab equipment including different panels, switches, push buttons, CBs, their locations, and functions, for thorough understanding to perform tasks independently	-	-	-	-
PC13. inspect and assess bogie and wheels, gangway, and coupler systems by identifying different parts and equipment, along with their purpose and function	-	-	-	-
PC14. identify different types of brakes and their respective purposes and functions, including various pneumatic valves, compressors, reservoirs, and colour-coded piping	-	-	-	-
PC15. identify different doors types, their purposes and functions, and the location of equipment associated with doors	-	-	-	-
PC16. train on the saloon and cab AC, identifying major components, explaining the refrigeration cycle, and applying different modes of operation and temperature settings	-	-	-	-
PC17. operate the propulsion system, covering pantograph (Panto), vacuum circuit breaker (VCB), main transformer (MTR), control inverter (CI), and traction motors (TM)	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC18. carry out operational and functional checks on the auxiliary power supply system, covering major components within the Static Inverter (SIV) and battery systems	-	-	-	-
PC19. train on the TIMS and PA/PIS System, handling key operations, including modules like the AOP, MOP, PAMP, displays, and push buttons	-	-	-	-
Practical Training on other Rolling Stock System & Depot	**1**	**1**	**-**	**0.5**
PC20. train on saloon and cab equipment in other rolling stock stocks, covering different panels, switches, push buttons, and CBS, identifying their locations, functions and differences compared to rolling stock	-	-	-	-
PC21. inspect other rolling stock bogie and wheels, gangway, coupler, brakes, and pneumatic systems, applying practical knowledge of different parts, equipment functions, and differences from rolling stock	-	-	-	-
PC22. inspect other rolling stock door systems, identifying various door types, their purposes, functions, and associated equipment locations, while understanding operational differences from rolling stock	-	-	-	-
PC23. train on other rolling stock saloon and cab ac units, applying knowledge of major components, the refrigeration cycle, and temperature setting modes with noted differences from rolling stock	-	-	-	-
PC24. train on TIMS and PA/PIS in other rolling stock, handling key modules and operations such as displays, push buttons, and announcements, focusing on the differences from rolling stock systems	-	-	-	-
PC25. operate the other rolling stock propulsion system, including Panto, VCB, MTR, CI, and traction motors, ensuring familiarity with each component's location, function, and any differences from rolling stock	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC26. carry out operational checks on other rolling stock auxiliary power supply systems, including SIV and battery systems, identifying different loads, modules, and components, noting differences from rolling stock	-	-	-	-
Tools and Test Equipment	2	2	-	0.5
PC27. identify and select appropriate tools and test equipment based on their type, function, parameters, and purpose, ensuring their accurate application for maintenance and testing tasks in line with specific requirements	-	-	-	-
PC28. calibrate, and utilize tools and test equipment following manufacturer guidelines, technical standards, and established calibration procedures to ensure proper handling, reliability, and accuracy	-	-	-	-
Machinery and Plant (M&P)	2	2	-	0.5
PC29. identify, inspect, and operate depot M&P equipment, ensuring readiness, efficiency, and safe usage for maintenance tasks	-	-	-	-
PC30. perform routine checks on depot M&P and report issues promptly to ensure smooth operations and minimize downtime	-	-	-	-
Stores	1	1	-	0.5
PC31. oversee custody stores, tool rooms, and DCOS, managing stock and non-stock items, consumables, and tools by maintaining organized records, overseeing issuance and returns, and ensuring operational readiness	-	-	-	-
PC32. forecast and monitor inventory requirements, including EAC and PAC, by conducting stock verification, implementing rotation practices, auditing compliance, and tracking budgets to minimize wastage and ensure efficient resource utilization	-	-	-	-
Safety Precautions	1	1	-	0.5

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC33. implement and enforce safety protocols in the Depot Yard, IBL, SBL, on roofs, near M&Ps, and OHE areas, including use of PPEs for various tasks	-	-	-	-
PC34. apply specific safety measures when working on trains, welding, and electrical maintenance, ensuring proper isolation, grounding, and protective gear	-	-	-	-
PC35. follow safe procedures for train coupling/uncoupling and shunting, adhering to alignment, signaling, and safe distance guidelines	-	-	-	-
PC36. execute safety checks for lifting operations and movement of heavy items, including use of overhead cranes, lifting gears, and proper techniques	-	-	-	-
PC37. ensure compliance with M&P safety protocols for equipment like Pit Jacks, Mobile Jacks, PWL, AWP, and RRV, covering secure setup and load limits	-	-	-	-
Maintenance Philosophy	**1**	**1**	**-**	**0.5**
PC38. distinguish between preventive, corrective, and breakdown maintenance, applying each as appropriate	-	-	-	-
PC39. document and report on maintenance activities, ensuring efficient resource use	-	-	-	-
Budget Management	**1**	**1**	**-**	**0.5**
PC40. prepare and manage budgets for rolling stock maintenance, including allocation for consumables and stock items	-	-	-	-
PC41. monitor budget utilization and optimize expenditures in rolling stock maintenance	-	-	-	-
Organizational Chart	**1**	**1**	**-**	**0.5**
PC42. familiarize staff with the organization chart and functional roles in the RS depot and CPTD Cell	-	-	-	-
PC43. facilitate collaboration among sections such as PPIO, INSP, TECH CELL, and FIT	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
21-Hour Reports and T&Q Cell Reporting	2	2	-	0.5
PC44. analyze 21-hour failure reports and other T&Q Cell documents to identify trends and recurring failures	-	-	-	-
PC45. analyze 21-hour failure reports and other T&Q Cell documents to identify trends and recurring failures	-	-	-	-
Section Responsibilities	1	1	-	0.5
PC46. familiarize with roles and responsibilities of sections like PPIO, INSP, TECH CELL, FIT, MW, RSC, and DCC	-	-	-	-
PC47. coordinate efficient functioning within and between sections for seamless RS operations	-	-	-	-
Rolling Stock Comparison	1	1	-	1
PC48. distinguish between all rolling stock types, identifying unique features and maintenance requirements	-	-	-	-
PC49. train the relevant personnel on stock-specific maintenance practices and equipment handling protocols	-	-	-	-
Depot & M&Ps	1	1	-	1
PC50. train on rolling stock depot layout, covering the depot's design, functional zones, and workflow to ensure effective navigation and operational efficiency within the depot environment	-	-	-	-
PC51. train on important M&Ps in rolling stock depots, including operation, maintenance procedures, and safety protocols specific to critical M&P equipment	-	-	-	-
Depot Control Center (DCC)	1	1	-	1
PC52. analyze the DCC working to understand the full operational flow, including monitoring, coordination, and communication functions essential for depot operations	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC53. determine the role and tasks of the Points man within the DCC framework, e.g. managing track switches, ensuring safe movement of trains within the depot, etc	-	-	-	-
Rolling Stock Care (RSC)	**1**	**1**	**-**	**1**
PC54. conduct RSC visits to inspect and ensure proper rolling stock care	-	-	-	-
PC55. report on RSC findings and coordinate with depot staff to address any identified issues	-	-	-	-
NOS Total	**20**	**20**	**-**	**10**

National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N0301
NOS Name	Understand and Compare Rolling Stock and Other Rolling Stock Systems,Safety, and M &P Fundamentals
Sector	Logistics
Sub-Sector	
Occupation	Rolling Stock
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

DMR/LSC/N0302: Develop Proficiency in Other Rolling Stock Systems

Description

This unit covers the technical aspects and operational differences across other rolling stocks systems, focusing on key functionalities, control mechanisms, and system-specific training essential for effective operation and maintenance.

Scope

The scope covers the following :

- Other Rolling Stocks System Training
- Training on Other Rolling Stock System & Depot
- Training on Rapid Metro Trains
- Training on Airport Line Trains

Elements and Performance Criteria

Other Rolling Stock System Training

To be competent, the user/individual on the job must be able to:

PC1. identify equipment layout, saloon interior & exterior, cab equipment overview, different panels, switches, push buttons, CBs locations, and functions of other Rolling Stocks, and changes in other Rolling Stocks

PC2. check equipment layout, saloon interior & exterior, cab equipment overview, different panels, switches, push buttons, CBs locations, and functions of other Rolling Stock

PC3. identify different TCMS modules fitted in different cars with their redundancy, and different modes/screens on Video Display Units of other Rolling Stock

PC4. identify different bogies & wheels parts, equipment, their purpose and function in other Rolling Stock, and changes in other Rolling Stock

PC5. identify different gangway & coupler parts, the purpose and function of equipment in other Rolling Stock, and changes in other Rolling Stock

PC6. identify different brakes & pneumatic system, including the brakes types, purpose and function, pneumatic valves, compressors, reservoirs, and piping with colour codes of other Rolling Stock

PC7. identify different types of doors in the door system, their purpose and functions, and the location of various equipment associated with doors of other Rolling Stock, and changes in other Rolling Stock

PC8. determine the purpose and functions of major Saloon & Cab AC components, refrigeration cycle, different operation modes of saloon & cab AC, temperature settings of other Rolling Stock, etc.

PC9. identify the other Rolling Stock propulsion system (Panto, VCB, MTR, CI & TM), and differences in other Rolling Stock

PC10. identify different Auxiliary Power Supply System in other Rolling Stock modules/components fitted in SIV & BATTERY with their location and function, and differences in other Rolling Stock

PC11. identify major PA/PIS systems, subsystems, and modules like AOP, MOP, PAMP, displays, push buttons, etc., of other Rolling Stock, and changes in other Rolling Stock

PC12. identify different TCMS modules fitted in different cars with their redundancy, and different modes/screens on Video Display Units of other Rolling Stock

Training on other Rolling Stock System & Depot

To be competent, the user/individual on the job must be able to:

PC13. identify cab equipment, including different panels, switches, push buttons, CBs locations, and functions

PC14. analyze the purpose and function of bogie & wheels, gangway & coupler, and brakes & pneumatic system

PC15. determine the purpose, location, and function of door system and Aircon (Saloon & Cab AC)

PC16. determine the role, functionality, and operation of TCMS & PAPIS within the other rolling stock system

PC17. analyze the functions, locations, and operations of Propulsion System (Panto, VCB, MTR, CI & TM), and Auxiliary Power Supply System (SIV & BATTERY)

PC18. identify the key areas, equipment, and overall layout of the other rolling stock depot

PC19. analyze the function and operational procedures for important M&Ps in other rolling stock Depots machines and piece of equipment

PC20. analyze the use, maintenance, and safety procedures associated with each M&P area and piece of equipment in the other rolling stock depot workshop

PC21. identify different cab equipment, panels, switches, push buttons, CBs locations, and functions specific to other rolling stock

PC22. determine the purpose and function of each bogie & wheels, gangway & coupler component on within the other rolling stock system

PC23. identify different types of brakes, pneumatic valves, compressors, reservoirs, and piping color codes in other rolling stock brake & pneumatic system

PC24. determine the purpose, location, and function of each component associated with the doors within other rolling stock door system

PC25. identify the major components, the refrigeration cycle, modes of operation, and temperature settings specific to other rolling stock saloon & cab AC

PC26. determine the role, functionality, and operations of TCMS & PAPIS within other rolling stock

PC27. determine the purpose, function, and operational process for the propulsion System (Panto, VCB, MTR, CI & TM) components within other rolling stock

PC28. determine the locations, purpose, and functions of each Auxiliary Power Supply System (SIV & BATTERY) component in other rolling stock

Training on Rapid Metro Trains

To be competent, the user/individual on the job must be able to:

PC29. identify different Saloon & Cab Equipment panels, switches, push buttons, CBs locations, and functions specific to rapid metro trains

PC30. determine the purpose and function of each bogie & wheel, gangway & coupler component within the rapid metro system

PC31. identify different types of brakes, pneumatic valves, compressors, reservoirs, and piping colour codes in Rapid Metro brakes & pneumatic system

PC32. determine the purpose, location, and function of each component associated with doors in rapid metro door system

PC33. identify the major components, refrigeration cycle, modes of operation, and temperature settings specific to rapid metro trains saloon & cab AC

PC34. determine the roles, functionality, and operation of TIMS & PAPIS within rapid metro

PC35. identify the DC equipment and determine the DC power collection processes in rapid metro

PC36. determine the purpose, function, and operational process of the propulsion system components in rapid metro

PC37. determine the locations, purpose, and functions of auxiliary power supply system components in rapid metro

Training on Airport Line Trains

To be competent, the user/individual on the job must be able to:

PC38. identify different saloon & cab equipment panels, switches, push buttons, CBs locations, and functions specific to airport line trains

PC39. determine the purpose, components, and functions relevant to airport line trains bogie & wheels, gangway & coupler

PC40. identify the types of brakes, pneumatic valves, compressors, reservoirs, and colour-coded piping specific to airport line train brakes & pneumatic system

PC41. determine the purpose, location, and function of each component associated with doors in airport line train door system

PC42. identify the major components, refrigeration cycle, modes of operation, and temperature settings of airport line train saloon & cab AC

PC43. determine the roles, functionality, and operation of TIMS & PAPIS systems specific to other rolling stock used on airport line

PC44. identify the function and location of Panto, VCB, MTR, CI, and TM in airport line train propulsion system

PC45. determine the purpose, function, and location of Auxiliary Power Supply System components within the airport line trains, including the SIV and BATTERY

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the overview and layout of Other Rolling Stock, detailing interior/exterior saloon and cab equipment, panels, switches, push buttons, CBs, and functions

KU2. Other Rolling Stock differences in equipment, layouts, and functionalities

KU3. the overview of TCMS in Other Rolling Stock, including modules with redundancy and Video Display Unit (VDU) modes/screen

KU4. TCMS in Other Rolling Stock and comparison with Other Rolling Stock

KU5. the overview of bogies, wheels, and couplers in Other Rolling Stock, detailing parts, equipment, purposes, and functions, and the differences in bogies, wheels, and couplers for Other Rolling Stock

KU6. the overview of Other Rolling Stock brake and pneumatic systems, types of brakes, valves, compressors, reservoirs, piping, and colour coding

KU7. the overview of Other Rolling Stock door systems, door types, purposes, functions, and equipment locations and the variations in Other Rolling Stock door systems

KU8. the Other Rolling Stock air conditioning overview, including main components, refrigeration cycle, operational modes, and temperature settings, and the modifications in Other Rolling Stock air conditioning

KU9. the propulsion system overview for Other Rolling Stock, including main elements and Other Rolling Stock differences

KU10. the overview of Other Rolling Stock auxiliary power supply (SIV & Battery), components, loads, and ratings and Other Rolling Stock system variations

KU11. the overview of Other Rolling Stock PA/PIS, including AOP, MOP, PAMP, displays, push buttons, and modifications in Other Rolling Stock

KU12. the Other Rolling Stocks cab equipment & panels, including cab layout, panels, switches, push buttons, CBs, and functions

KU13. the Other Rolling Stocks bogie, wheels, gangway, coupler, brakes & pneumatic system, including the structure, functions, and safety

KU14. the Other Rolling Stocks door system and air conditioning systems

KU15. Other Rolling Stock TCMS, VDUs, and PA/PIS systems

KU16. the propulsion system, SIV, and battery systems for Other Rolling Stock

KU17. Other Rolling Stock depot infrastructure and maintenance procedures

KU18. M&Ps in Other Rolling Stock Depots & Workshop

KU19. TIMS and PA/PIS of Other Rolling Stocks

KU20. the Airport Line cab and saloon layouts, equipment, panels, and systems

KU21. the bogies, wheels, gangways, couplers, brakes, pneumatic systems, doors, and air conditioning specific to the Airport Line

KU22. DC power collection methods, propulsion systems, and auxiliary power operations

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. maintain detailed and accurate records of system checks, equipment identification, and maintenance activities related to Other Rolling Stock systems

GS2. communicate clearly and effectively with team members, junior engineers, and other departments regarding the status, issues, and maintenance requirements of Other Rolling Stock systems

GS3. prepare reports and updates on equipment condition, failures, or preventive maintenance tasks

GS4. coordinate with relevant departments, including maintenance, operations, and depot management, to ensure smooth functioning and timely repairs of Other Rolling Stock components

GS5. apply problem-solving skills to diagnose issues with Other Rolling Stock systems, such as electrical failures, pneumatic issues, or propulsion system malfunctions

GS6. use logical thinking to analyze system behavior, identify root causes, and recommend appropriate corrective actions

GS7. make decisions regarding the prioritization of maintenance tasks for Other Rolling Stock systems, considering the criticality of components like brakes, propulsion systems, and auxiliary power supplies

GS8. ensure timely escalation of complex issues to higher levels of management

GS9. prioritize tasks based on the severity of issues to ensure efficient use of resources and minimal downtime

GS10. follow safety protocols when handling Other Rolling Stock equipment and ensure all safety procedures are adhered to during maintenance and troubleshooting activities

GS11. work effectively as part of a team, sharing knowledge of Other Rolling Stock systems with colleagues and supporting each other in problem resolution

GS12. adapt to changes in the Other Rolling Stock systems, such as upgrades or modifications, and adapt to new maintenance practices, technological changes, or emergencies that may arise

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Other Rolling Stock System Training	**4**	**4**	**-**	**2**
PC1. identify equipment layout, saloon interior & exterior, cab equipment overview, different panels, switches, push buttons, CBs locations, and functions of other Rolling Stocks, and changes in other Rolling Stocks	-	-	-	-
PC2. check equipment layout, saloon interior & exterior, cab equipment overview, different panels, switches, push buttons, CBs locations, and functions of other Rolling Stock	-	-	-	-
PC3. identify different TCMS modules fitted in different cars with their redundancy, and different modes/screens on Video Display Units of other Rolling Stock	-	-	-	-
PC4. identify different bogies & wheels parts, equipment, their purpose and function in other Rolling Stock, and changes in other Rolling Stock	-	-	-	-
PC5. identify different gangway & coupler parts, the purpose and function of equipment in other Rolling Stock, and changes in other Rolling Stock	-	-	-	-
PC6. identify different brakes & pneumatic system, including the brakes types, purpose and function, pneumatic valves, compressors, reservoirs, and piping with colour codes of other Rolling Stock	-	-	-	-
PC7. identify different types of doors in the door system, their purpose and functions, and the location of various equipment associated with doors of other Rolling Stock, and changes in other Rolling Stock	-	-	-	-
PC8. determine the purpose and functions of major Saloon & Cab AC components, refrigeration cycle, different operation modes of saloon & cab AC, temperature settings of other Rolling Stock, etc.	-	-	-	-
PC9. identify the other Rolling Stock propulsion system (Panto, VCB, MTR, CI & TM), and differences in other Rolling Stock	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. identify different Auxiliary Power Supply System in other Rolling Stock modules/components fitted in SIV & BATTERY with their location and function, and differences in other Rolling Stock	-	-	-	-
PC11. identify major PA/PIS systems, subsystems, and modules like AOP, MOP, PAMP, displays, push buttons, etc., of other Rolling Stock, and changes in other Rolling Stock	-	-	-	-
PC12. identify different TCMS modules fitted in different cars with their redundancy, and different modes/screens on Video Display Units of other Rolling Stock	-	-	-	-
Training on other Rolling Stock System & Depot	**8**	**8**	**-**	**4**
PC13. identify cab equipment, including different panels, switches, push buttons, CBs locations, and functions	-	-	-	-
PC14. analyze the purpose and function of bogie & wheels, gangway & coupler, and brakes & pneumatic system	-	-	-	-
PC15. determine the purpose, location, and function of door system and Aircon (Saloon & Cab AC)	-	-	-	-
PC16. determine the role, functionality, and operation of TCMS & PAPIS within the other rolling stock system	-	-	-	-
PC17. analyze the functions, locations, and operations of Propulsion System (Panto, VCB, MTR, CI & TM), and Auxiliary Power Supply System (SIV & BATTERY)	-	-	-	-
PC18. identify the key areas, equipment, and overall layout of the other rolling stock depot	-	-	-	-
PC19. analyze the function and operational procedures for important M&Ps in other rolling stock Depots machines and piece of equipment	-	-	-	-
PC20. analyze the use, maintenance, and safety procedures associated with each M&P area and piece of equipment in the other rolling stock depot workshop	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC21. identify different cab equipment, panels, switches, push buttons, CBs locations, and functions specific to other rolling stock	-	-	-	-
PC22. determine the purpose and function of each bogie & wheels, gangway & coupler component on within the other rolling stock system	-	-	-	-
PC23. identify different types of brakes, pneumatic valves, compressors, reservoirs, and piping color codes in other rolling stock brake & pneumatic system	-	-	-	-
PC24. determine the purpose, location, and function of each component associated with the doors within other rolling stock door system	-	-	-	-
PC25. identify the major components, the refrigeration cycle, modes of operation, and temperature settings specific to other rolling stock saloon & cab AC	-	-	-	-
PC26. determine the role, functionality, and operations of TCMS & PAPIS within other rolling stock	-	-	-	-
PC27. determine the purpose, function, and operational process for the propulsion System (Panto, VCB, MTR, CI & TM) components within other rolling stock	-	-	-	-
PC28. determine the locations, purpose, and functions of each Auxiliary Power Supply System (SIV & BATTERY) component in other rolling stock	-	-	-	-
Training on Rapid Metro Trains	**4**	**4**	**-**	**2**
PC29. identify different Saloon & Cab Equipment panels, switches, push buttons, CBs locations, and functions specific to rapid metro trains	-	-	-	-
PC30. determine the purpose and function of each bogie & wheel, gangway & coupler component within the rapid metro system	-	-	-	-
PC31. identify different types of brakes, pneumatic valves, compressors, reservoirs, and piping colour codes in Rapid Metro brakes & pneumatic system	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC32. determine the purpose, location, and function of each component associated with doors in rapid metro door system	-	-	-	-
PC33. identify the major components, refrigeration cycle, modes of operation, and temperature settings specific to rapid metro trains saloon & cab AC	-	-	-	-
PC34. determine the roles, functionality, and operation of TIMS & PAPIS within rapid metro	-	-	-	-
PC35. identify the DC equipment and determine the DC power collection processes in rapid metro	-	-	-	-
PC36. determine the purpose, function, and operational process of the propulsion system components in rapid metro	-	-	-	-
PC37. determine the locations, purpose, and functions of auxiliary power supply system components in rapid metro	-	-	-	-
Training on Airport Line Trains	**4**	**4**	**-**	**2**
PC38. identify different saloon & cab equipment panels, switches, push buttons, CBs locations, and functions specific to airport line trains	-	-	-	-
PC39. determine the purpose, components, and functions relevant to airport line trains bogie & wheels, gangway & coupler	-	-	-	-
PC40. identify the types of brakes, pneumatic valves, compressors, reservoirs, and colour-coded piping specific to airport line train brakes & pneumatic system	-	-	-	-
PC41. determine the purpose, location, and function of each component associated with doors in airport line train door system	-	-	-	-
PC42. identify the major components, refrigeration cycle, modes of operation, and temperature settings of airport line train saloon & cab AC	-	-	-	-
PC43. determine the roles, functionality, and operation of TIMS & PAPIS systems specific to other rolling stock used on airport line	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC44. identify the function and location of Panto, VCB, MTR, CI, and TM in airport line train propulsion system	-	-	-	-
PC45. determine the purpose, function, and location of Auxiliary Power Supply System components within the airport line trains, including the SIV and BATTERY	-	-	-	-
NOS Total	**20**	**20**	**-**	**10**

National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N0302
NOS Name	Develop Proficiency in Other Rolling Stock Systems
Sector	Logistics
Sub-Sector	
Occupation	Rolling Stock
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

DMR/LSC/N0303: Implement Key Safety and Maintenance Procedures for Rolling Stock

Description

This unit covers critical maintenance procedures and safety protocols necessary for ensuring the safe, reliable, and efficient operation of rolling stock, emphasizing preventive measures and best practices to reduce risks during maintenance activities.

Scope

The scope covers the following :

- Safety precautions
- Shunting procedure
- Coupling procedure
- Rescue procedure
- Lifting appliance use and rerailing
- Axle lock

Elements and Performance Criteria

Safety precautions

To be competent, the user/individual on the job must be able to:

PC1. follow the procedure for issuing a job card, ensuring all necessary details are documented and the job is assigned to the appropriate personnel

PC2. execute the procedure to issue a train fitness certificate for revenue service, verifying all safety and operational checks are complete before approving for service

PC3. adhere to the procedure for issuing a power block, ensuring proper authorization and documentation to safely de-energize and clear the track section for maintenance work

PC4. follow the procedure for energization of a power block in the inspection bay, ensuring safe restoration of power once maintenance work is completed

PC5. follow the procedure for issuing and canceling a power block for Retractable OHE, ensuring compliance with safety measures during both issuance and cancellation after work completion

PC6. implement the appropriate procedure for safe working on the train, including isolation procedures and use of necessary safety equipment

PC7. implement the procedure for working in the under gear of the train on the IBL, ensuring safe access to and handling of undercarriage components with appropriate safety protocols

PC8. adhere to the procedure for maintenance work in the depot yard, following specific yard safety protocols and guidelines for safe movement and operation in the area

PC9. conduct Safety Integrity Check (SIC) after first-line PM/CM work, ensuring all preventive or corrective maintenance tasks are completed to safety standards before the train operations

PC10. carry out preventive checks on the train, ensuring thorough inspection and identification of potential issues to prevent operational disruptions or safety hazards

Shunting procedure

To be competent, the user/individual on the job must be able to:

PC11. oversee the shunting of non-powered or defective train in the depot, ensuring adherence to safety protocols and appropriate methods to safely maneuver the train without self-propulsion

PC12. follow the procedure for shunting a self-powered train in the depot, including guidelines for operating the train’s own propulsion system to safely move it within depot boundaries

PC13. execute the procedure for stabling an uncoupled train, confirming the train is properly secured and positioned to prevent any unintended movement

PC14. oversee the stabling of defective trains, ensuring all necessary safety measures are taken, e.g. applying handbrakes, securing the location, and alerting relevant personnel

PC15. implement the procedure for changing brake blocks on the train, ensuring proper safety precautions, correct tool use, and accurate replacement to maintain optimal braking performance

PC16. implement guidelines concerning mobile phones usage in the depot premises, ensuring it does not interfere with operational safety, especially during active shunting or maintenance tasks

PC17. follow the procedure for testing the train on the test track, including necessary pre-test inspections, safe operation protocols, and post-test evaluations

PC18. oversee shunting of trains on the test track, adhering to specific safety measures to safely conduct shunting maneuvers within the test environment

Coupling procedure

To be competent, the user/individual on the job must be able to:

PC19. oversee the coupling and uncoupling of trains on a unit basis, ensuring each unit is securely connected or disconnected following applicable safety protocols and operational guidelines

PC20. check coupling and uncoupling (M-M) and (DT-M) configurations, ensuring units are safely joined or separated according to the type of train formation and applicable standards

PC21. carry out automatic coupling/ uncoupling of two units (M-M) of other rolling stock trains, operating the automatic system effectively for safe coupling and uncoupling of motor cars

PC22. perform uncoupling and coupling of M-M car gangway on other rolling stock, ensuring secure connection and disconnection of gangways to maintain continuity and safe car movement

PC23. execute hostler operation for movement of units in other rolling stock, ensuring safe handling of the train’s hostler system to maneuver units within the depot for positioning or maintenance

Rescue procedure

To be competent, the user/individual on the job must be able to:

PC24. conduct the rescue of rolling stock disabled trains on the mainline using another rolling stock train, ensuring safe and effective coupling, movement, and transportation back to the depot

PC25. execute the rescue of other rolling stock disabled trains on the mainline using another other rolling stock train, verifying correct alignment, secure coupling, and communication between trains for safe rescue

PC26. follow the procedure for pulling a defective train with a locomotive, conducting necessary checks and maintaining appropriate speed limits for safe transport

PC27. oversee pushing of defective trains with locomotive, ensuring adherence to safety protocols and coordination steps to prevent derailment and steady movement to the destination

PC28. oversee the rescue of disabled trains on the mainline using another rolling stock train, ensuring compatibility, secure connections, and stability throughout the operation to prevent accidents

PC29. handle pantograph entanglement with OHE on the mainline, safety isolating power, disentangling the pantograph, and ensuring minimal damage to the train and OHE system

Lifting appliance use and rerailing

To be competent, the user/individual on the job must be able to:

PC30. ensure the safe operation of lifting appliances and lifting gears by conducting pre-use inspections, verifying certifications, adhering to safety guidelines for load limits, securing the load properly, and using the access platform as per safety standards

PC31. execute heavy lifting operations using the overhead crane by following prescribed safety protocols, conducting a test lift to ensure load stability, using certified lifting slings, and ensuring precise and safe rerailing in the inspection bay

Axle lock

To be competent, the user/individual on the job must be able to:

PC32. ensure the safe movement of trains with axle lock on the main line and Rail Cum Road Vehicles (RRVs), adhering to alignment and transition procedures, and maintaining controlled speeds to prevent accidents or derailments

PC33. perform rerailing of DT or M car bogies by preparing the site, utilizing appropriate tools and equipment, realigning bogies onto the track with precision, and verifying operational stability for safe resumption of service

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the procedure for issuing a job card and train fitness certificate for revenue service

KU2. the procedure for issuing a power block and energizing a power block in the inspection bay

KU3. the procedure for issuance and cancellation of a power block for retractable Overhead Equipment (OHE)

KU4. the procedure for working on the train and working under the gear of a train on the Inspection Bay Line (IBL)

KU5. the procedure for maintenance work in the depot yard

KU6. Standard Inspection Certificate (SIC) issuance after first-line Preventive Maintenance (PM) or Corrective Maintenance (CM) work, and preventive checks on the train

KU7. the procedure for shunting a non-powered or defective train in the depot

KU8. the procedure for shunting a self-powered train in the depot

KU9. the procedure for stabling an uncoupled train

KU10. the procedure for stabling a defective train

KU11. the procedure for changing the brake block on the train and guidelines for using mobile phones in the depot premises

KU12. the procedure for testing the train on the test track

KU13. the procedure for shunting a train on the test track

KU14. the procedure for coupling and uncoupling trains on a unit basis

KU15. the procedure for coupling and uncoupling trains (Motor-Motor) and (Driving Trailer-Motor) and for automatic coupling/uncoupling of two units (Motor-Motor) on other rolling stock trains

KU16. the procedure for uncoupling and coupling Motor-Motor car gangways and moving a unit using hostler operation in other rolling stock

KU17. the procedure for rescuing a disabled rolling stock train on the mainline using another rolling stock train

KU18. the procedure for rescuing a disabled other rolling stock systems train on the mainline using another other rolling stock train

KU19. the procedure for pulling and pushing a defective train with a locomotive

KU20. the procedure for rescuing a disabled train on the mainline using another train (other rolling stock)

KU21. the procedure for handling pantograph entanglement with OHE on the mainline

KU22. the guidelines for the safe use of lifting appliances, lifting gears, and safety procedures for operating access platforms

KU23. the procedure for using an overhead crane for heavy lifting and for operating overhead cranes for heavy lifting in inspection bay lines

KU24. the procedure for moving a train in case of an axle lock on the mainline (Driving Trailer or Motor car)

KU25. the procedure for rerailing bogies of Driving Trailer (DT) or Motor cars

KU26. the procedure for the movement of a Rail Cum Road Vehicle (RRV) on the track or mainline

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. ensure accurate recording of job cards, safety checks, power block issuances, and maintenance tasks

GS2. communicate safety protocols, train statuses, and any issues with team members and junior engineers effectively

GS3. coordinate tasks with operations and maintenance teams to ensure safe and efficient train handling, rescues, and maintenance

GS4. identify and resolve issues during tasks like shunting, coupling, and rescues, ensuring minimal risk and safety

GS5. make timely, safe decisions regarding train movements, power block procedures, and emergency responses

GS6. prioritize and complete safety procedures, checks, and maintenance tasks within required time frames

GS7. adhere strictly to safety protocols in all procedures, including train handling, lifting, and emergency rescues

GS8. collaborate effectively with colleagues to ensure safety during maintenance, shunting, and emergency operations

GS9. adjust to changing conditions and unexpected issues, such as train breakdowns or safety hazards

GS10. follow updates on safety procedures and operational protocols to improve performance and safety outcomes

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Safety precautions	**2**	**2**	**-**	**1**
PC1. follow the procedure for issuing a job card, ensuring all necessary details are documented and the job is assigned to the appropriate personnel	-	-	-	-
PC2. execute the procedure to issue a train fitness certificate for revenue service, verifying all safety and operational checks are complete before approving for service	-	-	-	-
PC3. adhere to the procedure for issuing a power block, ensuring proper authorization and documentation to safely de-energize and clear the track section for maintenance work	-	-	-	-
PC4. follow the procedure for energization of a power block in the inspection bay, ensuring safe restoration of power once maintenance work is completed	-	-	-	-
PC5. follow the procedure for issuing and canceling a power block for Retractable OHE, ensuring compliance with safety measures during both issuance and cancellation after work completion	-	-	-	-
PC6. implement the appropriate procedure for safe working on the train, including isolation procedures and use of necessary safety equipment	-	-	-	-
PC7. implement the procedure for working in the under gear of the train on the IBL, ensuring safe access to and handling of undercarriage components with appropriate safety protocols	-	-	-	-
PC8. adhere to the procedure for maintenance work in the depot yard, following specific yard safety protocols and guidelines for safe movement and operation in the area	-	-	-	-
PC9. conduct Safety Integrity Check (SIC) after first-line PM/CM work, ensuring all preventive or corrective maintenance tasks are completed to safety standards before the train operations	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. carry out preventive checks on the train, ensuring thorough inspection and identification of potential issues to prevent operational disruptions or safety hazards	-	-	-	-
Shunting procedure	**4**	**4**	**-**	**3**
PC11. oversee the shunting of non-powered or defective train in the depot, ensuring adherence to safety protocols and appropriate methods to safely maneuver the train without self-propulsion	-	-	-	-
PC12. follow the procedure for shunting a self-powered train in the depot, including guidelines for operating the train’s own propulsion system to safely move it within depot boundaries	-	-	-	-
PC13. execute the procedure for stabling an uncoupled train, confirming the train is properly secured and positioned to prevent any unintended movement	-	-	-	-
PC14. oversee the stabling of defective trains, ensuring all necessary safety measures are taken, e.g. applying handbrakes, securing the location, and alerting relevant personnel	-	-	-	-
PC15. implement the procedure for changing brake blocks on the train, ensuring proper safety precautions, correct tool use, and accurate replacement to maintain optimal braking performance	-	-	-	-
PC16. implement guidelines concerning mobile phones usage in the depot premises, ensuring it does not interfere with operational safety, especially during active shunting or maintenance tasks	-	-	-	-
PC17. follow the procedure for testing the train on the test track, including necessary pre-test inspections, safe operation protocols, and post-test evaluations	-	-	-	-
PC18. oversee shunting of trains on the test track, adhering to specific safety measures to safely conduct shunting maneuvers within the test environment	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Coupling procedure	**4**	**4**	**-**	**2**
PC19. oversee the coupling and uncoupling of trains on a unit basis, ensuring each unit is securely connected or disconnected following applicable safety protocols and operational guidelines	-	-	-	-
PC20. check coupling and uncoupling (M-M) and (DT-M) configurations, ensuring units are safely joined or separated according to the type of train formation and applicable standards	-	-	-	-
PC21. carry out automatic coupling/ uncoupling of two units (M-M) of other rolling stock trains, operating the automatic system effectively for safe coupling and uncoupling of motor cars	-	-	-	-
PC22. perform uncoupling and coupling of M-M car gangway on other rolling stock, ensuring secure connection and disconnection of gangways to maintain continuity and safe car movement	-	-	-	-
PC23. execute hostler operation for movement of units in other rolling stock, ensuring safe handling of the train's hostler system to maneuver units within the depot for positioning or maintenance	-	-	-	-
Rescue procedure	**4**	**4**	**-**	**2**
PC24. conduct the rescue of rolling stock disabled trains on the mainline using another rolling stock train, ensuring safe and effective coupling, movement, and transportation back to the depot	-	-	-	-
PC25. execute the rescue of other rolling stock disabled trains on the mainline using another other rolling stock train, verifying correct alignment, secure coupling, and communication between trains for safe rescue	-	-	-	-
PC26. follow the procedure for pulling a defective train with a locomotive, conducting necessary checks and maintaining appropriate speed limits for safe transport	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. oversee pushing of defective trains with locomotive, ensuring adherence to safety protocols and coordination steps to prevent derailment and steady movement to the destination	-	-	-	-
PC28. oversee the rescue of disabled trains on the mainline using another rolling stock train, ensuring compatibility, secure connections, and stability throughout the operation to prevent accidents	-	-	-	-
PC29. handle pantograph entanglement with OHE on the mainline, safety isolating power, disentangling the pantograph, and ensuring minimal damage to the train and OHE system	-	-	-	-
Lifting appliance use and rerailing	**3**	**3**	**-**	**1**
PC30. ensure the safe operation of lifting appliances and lifting gears by conducting pre-use inspections, verifying certifications, adhering to safety guidelines for load limits, securing the load properly, and using the access platform as per safety standards	-	-	-	-
PC31. execute heavy lifting operations using the overhead crane by following prescribed safety protocols, conducting a test lift to ensure load stability, using certified lifting slings, and ensuring precise and safe rerailing in the inspection bay	-	-	-	-
Axle lock	**3**	**3**	**-**	**1**
PC32. ensure the safe movement of trains with axle lock on the main line and Rail Cum Road Vehicles (RRVs), adhering to alignment and transition procedures, and maintaining controlled speeds to prevent accidents or derailments	-	-	-	-
PC33. perform rerailing of DT or M car bogies by preparing the site, utilizing appropriate tools and equipment, realigning bogies onto the track with precision, and verifying operational stability for safe resumption of service	-	-	-	-
NOS Total	**20**	**20**	**-**	**10**

Logo Not Available

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N0303
NOS Name	Implement Key Safety and Maintenance Procedures for Rolling Stock
Sector	Logistics
Sub-Sector	
Occupation	Rolling Stock
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

DMR/LSC/N0304: Conduct Preventive Maintenance Checks for Rolling Stock

Description

This unit is about performing systematic preventive maintenance checks for rolling stock, focusing on identifying potential issues early, ensuring equipment reliability, and minimizing downtime through routine inspections and maintenance tasks.

Scope

The scope covers the following :

- Conducting A check
- Conducting B-1 check
- Conducting B-4 check
- Conducting B-8 check
- Troubleshooting of RS faults

Elements and Performance Criteria

Conducting A check

To be competent, the user/individual on the job must be able to:

PC1. perform maintenance activities required for A service checks, using appropriate tools and diagnostic equipment

PC2. conduct practical inspection of components during A service checks, and document the findings and maintenance activities

Conducting B-1 check

To be competent, the user/individual on the job must be able to:

PC3. perform maintenance activities required for B-1 service checks, using appropriate tools and diagnostic equipment

PC4. conduct inspection of components during B-1 service checks, and document the findings and maintenance activities

Conducting B-4 check

To be competent, the user/individual on the job must be able to:

PC5. inspect key components during B-4 service check to identify wear or damage and ensure system functionality, adhering to applicable safety protocols

PC6. perform practical B-4 checks using appropriate tools, following the applicable procedures, and document the findings and maintenance activities

Conducting B-8 check

To be competent, the user/individual on the job must be able to:

PC7. perform maintenance activities as per the B-8 service check requirements

PC8. conduct practical inspections and testing as outlined for B-8 service checks, and document the findings and maintenance activities

Troubleshooting of RS faults

To be competent, the user/individual on the job must be able to:

PC9. identify faults in RS systems by interpreting diagnostic indicators and fault codes, and apply standard troubleshooting procedures to isolate root causes effectively

PC10. utilize appropriate tools and testing equipment to diagnose and repair faults, performing corrective actions safety, and confirm RS functionality through system checks

PC11. document troubleshooting activities, including fault identification, corrective measures, and results, in maintenance logs as per established standards

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. A service check maintenance requirements, including the specific components and systems to inspect, frequency of checks, and criteria for passing the check

KU2. the practical procedures for conducting A service checks, for inspecting, maintaining, and reporting on key components like brakes, wheels, electrical, pneumatic systems, and other safety-critical parts

KU3. the tools and equipment used during A service checks, including their proper usage, calibration requirements, and maintenance protocols to ensure accurate and effective diagnostics

KU4. the documentation protocols and compliance standards related to A service checks, to ensure all maintenance activities are recorded accurately and meet safety and quality standards

KU5. the scope and importance of B-1 service checks in the maintenance schedule

KU6. the tools and diagnostic equipment required for conducting B-1 checks

KU7. the standard operating procedures and protocols for B-1 check maintenance activities

KU8. the specific inspection criteria for components checked during B-1 service, such as brakes, electrical systems, and couplings

KU9. the safety protocols and Personal Protective Equipment (PPE) required during B-1 checks

KU10. the proper documentation practices and reporting requirements following a B-1 check

KU11. the signs of wear or failure in critical components evaluated in the B-1 check

KU12. the purpose and scope of B-4 service checks, including the specific components and systems that require inspection

KU13. the procedures and safety protocols for conducting B-4 maintenance activities, including tool and equipment usage and proper documentation

KU14. the standards for identifying and reporting wear, damage, or potential issues during a B-4 check to ensure equipment reliability and safety

KU15. the inspection and maintenance requirements specific to the B-8 service check

KU16. the use of specialized tools, gauges, and testing equipment necessary for B-8 service checks

KU17. the critical components and systems covered in B-8 checks, such as suspension, traction systems, and control panels

KU18. the safety guidelines and PPE requirements for conducting B-8 checks safely

KU19. the common issues or faults specific to the B-8 check interval and signs of wear or malfunction

KU20. the regulatory and manufacturer standards related to B-8 checks
KU21. the procedures for documenting and reporting B-8 check outcomes and maintenance needs
KU22. the common RS fault codes and their associated issues across various RS systems
KU23. the functionality of diagnostic tools and test equipment specific to RS troubleshooting
KU24. the standard troubleshooting processes and methods for each type of RS system (e.g. propulsion, braking, door systems)
KU25. the safety protocols and precautions for troubleshooting high-voltage and moving components in RS
KU26. the methods for analyzing diagnostic indicators to distinguish between related faults and symptoms
KU27. the component-specific troubleshooting techniques and manufacturer guidelines
KU28. the procedures for documenting fault resolutions and maintaining accurate maintenance records

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. document all inspection findings, maintenance activities, and troubleshooting steps following protocol for each service check and troubleshooting activity
GS2. convey maintenance findings, potential issues, and resolutions with team members to maintain smooth operations and prevent miscommunication
GS3. conduct meticulous inspections during A, B-1, B-4, and B-8 service checks to identify wear, damage, or potential failures in critical components, following each service's specific criteria
GS4. apply diagnostic tools and standard troubleshooting protocols to isolate and address faults in RS systems, ensuring an accurate and efficient resolution
GS5. follow all safety protocols and utilize PPE to minimize risk during maintenance checks and troubleshooting of high-voltage or moving components
GS6. use tools and diagnostic equipment correctly, including ensuring proper calibration and maintenance, to support accurate and safe service checks and troubleshooting
GS7. analyze diagnostic indicators and fault codes to differentiate between similar issues and determine appropriate corrective actions in RS system faults
GS8. make informed decisions on corrective actions based on diagnostic findings to resolve faults effectively while adhering to safety and operational standards
GS9. maintain compliance with all regulatory standards and manufacturer guidelines for each type of service check to ensure maintenance quality and operational reliability
GS10. identify and address wear, damage, or other issues in critical components, using knowledge of each service level's focus areas and procedures to ensure equipment reliability

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Conducting A check	**4**	**4**	**-**	**2**
PC1. perform maintenance activities required for A service checks, using appropriate tools and diagnostic equipment	-	-	-	-
PC2. conduct practical inspection of components during A service checks, and document the findings and maintenance activities	-	-	-	-
Conducting B-1 check	**4**	**4**	**-**	**2**
PC3. perform maintenance activities required for B-1 service checks, using appropriate tools and diagnostic equipment	-	-	-	-
PC4. conduct inspection of components during B-1 service checks, and document the findings and maintenance activities	-	-	-	-
Conducting B-4 check	**4**	**4**	**-**	**2**
PC5. inspect key components during B-4 service check to identify wear or damage and ensure system functionality, adhering to applicable safety protocols	-	-	-	-
PC6. perform practical B-4 checks using appropriate tools, following the applicable procedures, and document the findings and maintenance activities	-	-	-	-
Conducting B-8 check	**4**	**4**	**-**	**2**
PC7. perform maintenance activities as per the B-8 service check requirements	-	-	-	-
PC8. conduct practical inspections and testing as outlined for B-8 service checks, and document the findings and maintenance activities	-	-	-	-
Troubleshooting of RS faults	**4**	**4**	**-**	**2**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. identify faults in RS systems by interpreting diagnostic indicators and fault codes, and apply standard troubleshooting procedures to isolate root causes effectively	-	-	-	-
PC10. utilize appropriate tools and testing equipment to diagnose and repair faults, performing corrective actions safety, and confirm RS functionality through system checks	-	-	-	-
PC11. document troubleshooting activities, including fault identification, corrective measures, and results, in maintenance logs as per established standards	-	-	-	-
NOS Total	**20**	**20**	**-**	**10**

National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N0304
NOS Name	Conduct Preventive Maintenance Checks for Rolling Stock
Sector	Logistics
Sub-Sector	
Occupation	Rolling Stock
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

DMR/LSC/N0305: Undergo GMSM & MSM Maintenance Simulator Training

Description

This unit covers the use of General Maintenance Simulation Model (GMSM) and Maintenance Simulation Model (MSM) simulators to practice and enhance maintenance skills. It focuses on simulating real-world maintenance scenarios to improve troubleshooting, fault diagnosis, and maintenance techniques for rolling stock systems

Scope

The scope covers the following :

- General Maintenance Simulation Model (GMSM)
- Maintenance Simulation Model (MSM)

Elements and Performance Criteria

General Maintenance Simulation Model (GMSM)

To be competent, the user/individual on the job must be able to:

PC1. follow standard procedures for accessing and navigating the GMSM interface in other rolling stock

PC2. use GMSM to monitor and manage other rolling stock operational data accurately

PC3. identify and interpret critical system information displayed on the GMSM for other rolling stock

PC4. apply GMSM data for efficient fault detection, diagnostics, and system health checks

PC5. record and report findings from GMSM analysis as per standard protocols

PC6. respond to GMSM alerts and indicators in alignment with operational guidelines

Maintenance Simulation Model (MSM)

To be competent, the user/individual on the job must be able to:

PC7. use the other rolling stock Maintenance Simulator to practice identifying and resolving common system faults, operating the simulator according to safety and training protocols

PC8. navigate different modules and scenarios within the other rolling stock Maintenance Simulator effectively

PC9. complete simulated maintenance tasks on other rolling stock components, and apply learnings from the simulator to enhance on-field other rolling stock maintenance skills

PC10. document simulator-based training activities, including fault identification and resolution steps

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the functionalities and features of the GMSM system specific to other rolling stock

KU2. the navigation and layout of the GMSM interface for other rolling stock operations

KU3. various data displays and system status indicators within GMSM
KU4. the fault signals, alerts, and warnings presented by the GMSM
KU5. the standard data recording and reporting practices for findings from GMSM analysis
KU6. the safety precautions and operational standards when using GMSM for other rolling stock
KU7. the purpose and functionality of the other rolling stock Maintenance Simulator
KU8. the various maintenance and troubleshooting scenarios available in the simulator
KU9. the layout, controls, and features of the other rolling stock simulator interface
KU10. how simulator scenarios replicate real-life other rolling stock maintenance conditions
KU11. the procedures for documenting and reporting training activities completed in the simulator
KU12. the best practices for transferring skills learned from the simulator to real-world maintenance on other rolling stock

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. access and navigate the GMSM and other rolling stock Maintenance Simulator interfaces efficiently to monitor and manage operational data
GS2. interpret critical system information accurately, including alerts and indicators displayed on the GMSM, to assess other rolling stock system health
GS3. utilize GMSM data to identify and diagnose potential faults in other rolling stock systems, applying relevant data for accurate fault detection
GS4. record findings from GMSM analysis and simulator activities, following standard protocols to ensure traceable and compliant documentation
GS5. respond promptly to GMSM alerts and fault indicators in line with operational guidelines, maintaining system readiness and safety
GS6. follow safety and operational protocols when using GMSM and the other rolling stock simulator to ensure safe and effective training and system operation
GS7. apply knowledge gained from simulator scenarios to real-world other rolling stock maintenance, ensuring a seamless transition of skills
GS8. report analysis results and findings from GMSM and simulator training activities effectively, ensuring clear communication and understanding of system conditions

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
General Maintenance Simulation Model (GMSM)	**10**	**10**	**-**	**5**
PC1. follow standard procedures for accessing and navigating the GMSM interface in other rolling stock	-	-	-	-
PC2. use GMSM to monitor and manage other rolling stock operational data accurately	-	-	-	-
PC3. identify and interpret critical system information displayed on the GMSM for other rolling stock	-	-	-	-
PC4. apply GMSM data for efficient fault detection, diagnostics, and system health checks	-	-	-	-
PC5. record and report findings from GMSM analysis as per standard protocols	-	-	-	-
PC6. respond to GMSM alerts and indicators in alignment with operational guidelines	-	-	-	-
Maintenance Simulation Model (MSM)	**10**	**10**	**-**	**5**
PC7. use the other rolling stock Maintenance Simulator to practice identifying and resolving common system faults, operating the simulator according to safety and training protocols	-	-	-	-
PC8. navigate different modules and scenarios within the other rolling stock Maintenance Simulator effectively	-	-	-	-
PC9. complete simulated maintenance tasks on other rolling stock components, and apply learnings from the simulator to enhance on-field other rolling stock maintenance skills	-	-	-	-
PC10. document simulator-based training activities, including fault identification and resolution steps	-	-	-	-
NOS Total	**20**	**20**	**-**	**10**

National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N0305
NOS Name	Undergo GMSM & MSM Maintenance Simulator Training
Sector	Logistics
Sub-Sector	
Occupation	Rolling Stock
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

DMR/LSC/N0306: Undergo Training on Important VCC & Pneumatic Circuits of Rolling Stock and Other Rolling Stocks

Description

This unit is about training on the vital Vehicle Control Circuits (VCC) and pneumatic circuits in rolling stock and other rolling stock systems. It covers the design, operation, and troubleshooting of VCC and pneumatic systems across various stock models, ensuring an understanding of the systems' functionality, safety protocols, and maintenance procedures for effective operation and fault management.

Scope

The scope covers the following :

- Abbreviations
- Activation of Train Lines
- Controlling Cab & Energization of Train
- Pantograph Raising & Control Circuit, and Auxiliary Power System
- Door Loop & Circuits
- Emergency Brake Loop & Parking Brake
- Coupling & Uncoupling
- VCB Control
- CMSB & Brake Loop
- Traction System & Speed Control
- Brake Loop & Circuits
- Air Conditioning
- Communication System
- Pneumatic Circuit Part-1
- Pneumatic Circuit Part-2
- Basic Concept of Other Rolling Stock VCC
- Train Control and Management System (TCMS) / Other Rolling Stock
- Motoring & Braking Circuit of Other Rolling Stock
- Pneumatic Circuit of Other Rolling Stock

Elements and Performance Criteria

Abbreviations

To be competent, the user/individual on the job must be able to:

PC1. identify and explain key abbreviations and components within the VCC rolling stock system

PC2. familiarize and record the functional roles of each component as per rolling stock standards

Activation of Train Lines

To be competent, the user/individual on the job must be able to:

PC3. activate train lines and verify successful operation through circuits

PC4. conduct safety checks for each activated train line to confirm operational status

Controlling Cab & Energization of Train

To be competent, the user/individual on the job must be able to:

PC5. control and energize the cab and main train power through circuits

PC6. test energization sequences to ensure train readiness

Pantograph Raising & Control Circuit, and Auxiliary Power System

To be competent, the user/individual on the job must be able to:

PC7. operate pantograph raising functions and monitor control circuit for continuity

PC8. manage auxiliary power systems (3251 and 55), ensuring consistent power flow

Door Loop & Circuits

To be competent, the user/individual on the job must be able to:

PC9. test door loop functionality, verify operation across circuits

PC10. conduct regular checks and troubleshoot any door loop issues in real time

Emergency Brake Loop & Parking Brake

To be competent, the user/individual on the job must be able to:

PC11. operate emergency brake loop and ensure parking brake activation on circuits

PC12. inspect brake loop functionality and perform emergency stop tests

Coupling & Uncoupling

To be competent, the user/individual on the job must be able to:

PC13. perform coupling and uncoupling operations using circuit

PC14. inspect mechanical and electrical interfaces for coupling/uncoupling integrity

VCB Control

To be competent, the user/individual on the job must be able to:

PC15. control and test VCB (Vacuum Circuit Breaker) operations using circuits

PC16. monitor VCB response and verify correct engagement/disengagement

CMSB & Brake Loop

To be competent, the user/individual on the job must be able to:

PC17. operate CMSB (Control Module Signal Block) and test brake loop functionality

PC18. inspect CMSB output and ensure reliable braking control

Traction System & Speed Control

To be competent, the user/individual on the job must be able to:

PC19. operate traction systems and monitor speed control through circuits

PC20. test speed response and traction power under varying load conditions

Brake Loop & Circuits

To be competent, the user/individual on the job must be able to:

PC21. operate brake loop across circuits, ensuring consistent braking power

PC22. inspect and verify each circuit in the brake loop for full functionality

Air Conditioning

To be competent, the user/individual on the job must be able to:

PC23. operate and test air conditioning system through circuits

PC24. inspect airflow, temperature control, and overall cooling performance

Communication System

To be competent, the user/individual on the job must be able to:

PC25. test and monitor communication systems, including circuits

PC26. conduct site tests for all communication channels and ensure signal clarity

Pneumatic Circuit Part-1

To be competent, the user/individual on the job must be able to:

PC27. inspect pneumatic components, including pressure regulation devices, to ensure proper pressure, airflow, and optimal functionality in Part-1 circuits

PC28. conduct pressure tests, verify the responsiveness of valves and cylinders, and perform routine maintenance and basic troubleshooting to maintain consistent performance in Part-1 pneumatic lines

Pneumatic Circuit Part-2

To be competent, the user/individual on the job must be able to:

PC29. inspect, test, and calibrate pneumatic controls, pressure regulators, sensors, and actuators in Part-2 circuits to ensure efficient operation of brake systems and other powered functions

PC30. diagnose and resolve malfunctions in Part-2 pneumatic circuits, perform emergency shutdown procedures, and ensure compliance with safety protocols

Basic Concept of Other rolling stock VCC

To be competent, the user/individual on the job must be able to:

PC31. identify key components of the other rolling stock Vehicle Control Circuit (VCC), explain their roles, and perform system checks to monitor functionality and identify irregularities

PC32. conduct VCC diagnostics using standard tools to verify circuit integrity, response, and operational effectiveness

Train Control and Management System (TCMS) / other rolling Stock

To be competent, the user/individual on the job must be able to:

PC33. operate the TCMS to monitor and manage vehicle diagnostics in other rolling stock, configuring and testing settings for various operational scenarios

PC34. conduct routine system checks on the TCMS, diagnose real-time system alerts and faults, and ensure all modules are fully functional

Motoring & Braking Circuit of other rolling Stock

To be competent, the user/individual on the job must be able to:

PC35. inspect, test, and adjust motoring and braking circuits to ensure operational efficiency, including proper stopping distances under varying loads

PC36. diagnose and repair issues with motor drive components or braking systems, implementing safety measures during maintenance and checks

Pneumatic Circuit of other rolling Stock

To be competent, the user/individual on the job must be able to:

PC37. inspect and test pneumatic circuits, verifying the functionality of air compressors, reservoirs, valves, and pressure levels, identifying and rectifying faults such as leaks

PC38. perform maintenance on pneumatic components like air dryers, brake cylinders, and pressure regulators, conducting safety checks before and after maintenance to ensure circuit readiness

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. key abbreviations and definitions used in rolling stock and other rolling stock vehicle control systems
KU2. overview of rolling stock and other rolling stock vehicle control components, roles, and standard operating practices
KU3. methods for documenting component functions, including troubleshooting and maintenance records
KU4. train line activation procedures, including safety protocols and troubleshooting common activation issues
KU5. operations of critical circuits (e.g., 3219, 58, 3211, 3212) and power distribution across rolling stock and other rolling stock
KU6. the procedures for pantograph and auxiliary power circuit operations, including circuit flow and power checks
KU7. the safety protocols for energizing cab functions, pantographs, and auxiliary power systems
KU8. the control and testing procedures for door loop circuits, troubleshooting door loop issues, and safety requirements
KU9. the emergency and parking brake circuit functions, including detailed testing and safety impact
KU10. the procedures for safe coupling/uncoupling and related circuit knowledge
KU11. the operating procedures for VCB circuits
KU12. CMSB and brake loop components
KU13. traction and speed control circuits, including testing and speed management protocols
KU14. the safety and functionality checks for brake loop circuits and continuity verification
KU15. AC circuit operation, including temperature regulation circuits and troubleshooting AC malfunctions
KU16. overview of rolling stock and other rolling stock communication systems, including circuit functions and troubleshooting common issues
KU17. the structure, function, and layout of rolling stock and other rolling stock pneumatic systems, including primary and advanced components
KU18. the procedures for maintaining pneumatic system integrity and safety, with emphasis on leak detection and valve calibration
KU19. the pneumatic system pressure standards and tolerances in other rolling stock, and diagnostics for pressure/flow issues
KU20. the safety protocols and PPE requirements when handling high-pressure pneumatic systems
KU21. the overview of VCC in other rolling stock, including component roles, layout, and testing/maintenance procedures
KU22. the high-voltage safety precautions in VCC operations for other rolling stock
KU23. the structure and functionality of TCMS in other rolling stock, including data monitoring, system control, and operational procedures
KU24. the troubleshooting of TCMS, handling system errors, firmware updates, and safe operating protocols
KU25. the principles of other rolling stock motoring and braking operations, including power flow and braking dynamics
KU26. the component-specific details for motor drives, braking actuators, and other rolling stock control modules

KU27. the maintenance and troubleshooting methods for other rolling stock motoring and braking circuits

KU28. other rolling stock pneumatic circuit components like compressors, reservoirs, valves, and air dryers

KU29. the techniques for air flow and pressure level testing, including diagnostics equipment knowledge

KU30. the protocols for safe handling of both electrical and mechanical elements within other rolling stock systems

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. document component functions, maintenance activities, and any issues encountered within train control systems to ensure accurate and traceable records

GS2. clearly communicate technical issues and procedural information with team members, especially during tasks involving pneumatic circuits, cab control, and train line activations

GS3. coordinate with team members during system tests and activations, ensuring seamless operation and safety, especially during train energization and pneumatic maintenance

GS4. inspect circuits, monitor components, and verify pneumatic pressure and flow to ensure precision and prevent faults or failures in safety-critical areas

GS5. utilize troubleshooting techniques to resolve operational issues within braking systems, VCB control, and pneumatic circuits to maintain safe and efficient train operations

GS6. make prompt, informed decisions when encountering alerts or issues in TCMS, brake circuits, or traction systems, maintaining safety and continuity of operations

GS7. adhere to safety protocols and guidelines, when working with high-pressure pneumatic circuits, traction systems, and emergency brake loops

GS8. adjust to diverse operational tasks, from train line activation to pneumatic system maintenance, adapting skills to address a range of technical requirements

GS9. analyze circuit diagnostics, TCMS data, and VCC configurations to assess system health, predict potential issues, and optimize maintenance schedules

GS10. use the appropriate diagnostic tools, testing equipment, and control systems, to ensure efficient operation and compliance with maintenance standards

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Abbreviations	**1**	**1**	**-**	**0.5**
PC1. identify and explain key abbreviations and components within the VCC rolling stock system	-	-	-	-
PC2. familiarize and record the functional roles of each component as per rolling stock standards	-	-	-	-
Activation of Train Lines	**1**	**1**	**-**	**0.5**
PC3. activate train lines and verify successful operation through circuits	-	-	-	-
PC4. conduct safety checks for each activated train line to confirm operational status	-	-	-	-
Controlling Cab & Energization of Train	**1**	**1**	**-**	**0.5**
PC5. control and energize the cab and main train power through circuits	-	-	-	-
PC6. test energization sequences to ensure train readiness	-	-	-	-
Pantograph Raising & Control Circuit, and Auxiliary Power System	**1**	**1**	**-**	**0.5**
PC7. operate pantograph raising functions and monitor control circuit for continuity	-	-	-	-
PC8. manage auxiliary power systems (3251 and 55), ensuring consistent power flow	-	-	-	-
Door Loop & Circuits	**1**	**1**	**-**	**0.5**
PC9. test door loop functionality, verify operation across circuits	-	-	-	-
PC10. conduct regular checks and troubleshoot any door loop issues in real time	-	-	-	-
Emergency Brake Loop & Parking Brake	**1**	**1**	**-**	**0.5**
PC11. operate emergency brake loop and ensure parking brake activation on circuits	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. inspect brake loop functionality and perform emergency stop tests	-	-	-	-
Coupling & Uncoupling	**1**	**1**	**-**	**0.5**
PC13. perform coupling and uncoupling operations using circuit	-	-	-	-
PC14. inspect mechanical and electrical interfaces for coupling/uncoupling integrity	-	-	-	-
VCB Control	**1**	**1**	**-**	**0.5**
PC15. control and test VCB (Vacuum Circuit Breaker) operations using circuits	-	-	-	-
PC16. monitor VCB response and verify correct engagement/disengagement	-	-	-	-
CMSB & Brake Loop	**1**	**1**	**-**	**0.5**
PC17. operate CMSB (Control Module Signal Block) and test brake loop functionality	-	-	-	-
PC18. inspect CMSB output and ensure reliable braking control	-	-	-	-
Traction System & Speed Control	**1**	**1**	**-**	**0.5**
PC19. operate traction systems and monitor speed control through circuits	-	-	-	-
PC20. test speed response and traction power under varying load conditions	-	-	-	-
Brake Loop & Circuits	**1**	**1**	**-**	**0.5**
PC21. operate brake loop across circuits, ensuring consistent braking power	-	-	-	-
PC22. inspect and verify each circuit in the brake loop for full functionality	-	-	-	-
Air Conditioning	**1**	**1**	**-**	**0.5**
PC23. operate and test air conditioning system through circuits	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. inspect airflow, temperature control, and overall cooling performance	-	-	-	-
Communication System	**1**	**1**	**-**	**0.5**
PC25. test and monitor communication systems, including circuits	-	-	-	-
PC26. conduct site tests for all communication channels and ensure signal clarity	-	-	-	-
Pneumatic Circuit Part-1	**1**	**1**	**-**	**0.5**
PC27. inspect pneumatic components, including pressure regulation devices, to ensure proper pressure, airflow, and optimal functionality in Part-1 circuits	-	-	-	-
PC28. conduct pressure tests, verify the responsiveness of valves and cylinders, and perform routine maintenance and basic troubleshooting to maintain consistent performance in Part-1 pneumatic lines	-	-	-	-
Pneumatic Circuit Part-2	**1**	**1**	**-**	**0.5**
PC29. inspect, test, and calibrate pneumatic controls, pressure regulators, sensors, and actuators in Part-2 circuits to ensure efficient operation of brake systems and other powered functions	-	-	-	-
PC30. diagnose and resolve malfunctions in Part-2 pneumatic circuits, perform emergency shutdown procedures, and ensure compliance with safety protocols	-	-	-	-
Basic Concept of Other rolling stock VCC	**1**	**1**	**-**	**0.5**
PC31. identify key components of the other rolling stock Vehicle Control Circuit (VCC), explain their roles, and perform system checks to monitor functionality and identify irregularities	-	-	-	-
PC32. conduct VCC diagnostics using standard tools to verify circuit integrity, response, and operational effectiveness	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Train Control and Management System (TCMS) / other rolling Stock	**1**	**1**	**-**	**0.5**
PC33. operate the TCMS to monitor and manage vehicle diagnostics in other rolling stock, configuring and testing settings for various operational scenarios	-	-	-	-
PC34. conduct routine system checks on the TCMS, diagnose real-time system alerts and faults, and ensure all modules are fully functional	-	-	-	-
Motoring & Braking Circuit of other rolling Stock	**1**	**1**	**-**	**0.5**
PC35. inspect, test, and adjust motoring and braking circuits to ensure operational efficiency, including proper stopping distances under varying loads	-	-	-	-
PC36. diagnose and repair issues with motor drive components or braking systems, implementing safety measures during maintenance and checks	-	-	-	-
Pneumatic Circuit of other rolling Stock	**2**	**2**	**-**	**1**
PC37. inspect and test pneumatic circuits, verifying the functionality of air compressors, reservoirs, valves, and pressure levels, identifying and rectifying faults such as leaks	-	-	-	-
PC38. perform maintenance on pneumatic components like air dryers, brake cylinders, and pressure regulators, conducting safety checks before and after maintenance to ensure circuit readiness	-	-	-	-
NOS Total	**20**	**20**	**-**	**10**

National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N0306
NOS Name	Undergo Training on Important VCC & Pneumatic Circuits of Rolling Stock and Other Rolling Stocks
Sector	Logistics
Sub-Sector	
Occupation	Rolling Stock
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

KU10. how to compute income and expenses
KU11. importance of maintaining safety and security in financial transactions
KU12. different legal rights and laws
KU13. how to operate digital devices and applications safely and securely
KU14. ways to identify business opportunities
KU15. types of customers and their needs
KU16. how to apply for a job and prepare for an interview
KU17. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. communicate effectively using appropriate language
GS2. behave politely and appropriately with all
GS3. perform basic calculations
GS4. solve problems effectively
GS5. be careful and attentive at work
GS6. use time effectively
GS7. maintain hygiene and sanitisation to avoid infection

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Introduction to Employability Skills	**1**	**1**	**-**	**-**
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
Constitutional values – Citizenship	**1**	**1**	**-**	**-**
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
Becoming a Professional in the 21st Century	**1**	**3**	**-**	**-**
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
Basic English Skills	**2**	**3**	**-**	**-**
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
Communication Skills	**1**	**1**	**-**	**-**
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
Diversity & Inclusion	**1**	**1**	**-**	**-**
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
Financial and Legal Literacy	**3**	**4**	**-**	**-**
PC9. use various financial products and services safely and securely	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
Essential Digital Skills	**4**	**6**	**-**	**-**
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
Entrepreneurship	**3**	**5**	**-**	**-**
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
Customer Service	**2**	**2**	**-**	**-**
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
Getting ready for apprenticeship & Jobs	**1**	**3**	**-**	**-**
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	**20**	**30**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/09/2026
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.

2. The assessment for the theory part will be based on a knowledge bank of questions created by the SSC.

3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.

4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training centre (as per the assessment criteria below).

5. Individual assessment agencies will create unique evaluations for skill practicals for every student at each examination/training centre based on this criterion.

6. To pass the Qualification Pack, every trainee should score a minimum of 60% of aggregate marks to successfully clear the assessment.

7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 60

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
DMR/LSC/N9901.Implement metro operations training and safety procedures	20	20	0	10	50	10
DMR/LSC/N0301.Understand and Compare Rolling Stock and Other Rolling Stock Systems,Safety, and M &P Fundamentals	20	20	-	10	50	15
DMR/LSC/N0302.Develop Proficiency in Other Rolling Stock Systems	20	20	-	10	50	15
DMR/LSC/N0303.Implement Key Safety and Maintenance Procedures for Rolling Stock	20	20	-	10	50	15
DMR/LSC/N0304.Conduct Preventive Maintenance Checks for Rolling Stock	20	20	-	10	50	15
DMR/LSC/N0305.Undergo GMSM & MSM Maintenance Simulator Training	20	20	-	10	50	15
DMR/LSC/N0306.Undergo Training on Important VCC & Pneumatic Circuits of Rolling Stock and Other Rolling Stocks	20	20	-	10	50	10
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	5

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
Total	160	170	0	70	400	100

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NCVET	National Council for Vocational Education and Training
QP	Qualification Pack
MC	Model Curriculum
NSQC	National Skills Qualification Committee
NOS	National Occupational Standards
NCO	National Classification of Occupations
ES	Employability Skills
OJT	On the Job Training
RS	Rolling Stock
CB	Circuit Breaker
PA	Public Address
PIS	Passenger Information System
AOP	Automatic Operational Panel
MOP	Manual Operating Panel
PAMP	Passenger Announcement and Monitoring Panel
VCB	Vacuum Circuit Breaker
MTR	Main Transformer
CI	Current Interrupter
TM	Traction Motor
PT	Potential Transformer

CT	Current Transformer
EGS	Earth Grounding System
M&P	Machinery and Plant
OHE	Overhead Equipment
GMSM	General Maintenance Simulation Model
MSM	Maintenance Simulation Model
TIMS	Train Information and Management System
PWL	Platform Wheel Lifts
AWP	Aerial Work Platforms
RRV	Rail-Road Vehicles
CMSB	Control Module Signal Block
VCC	Vehicle Control Center
VCB	Vacuum Circuit Breaker

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.